Lehrstuhl für Data Science

Zero-Shot Narrative Classification via Agentic and Actor-Critic LLM Pipelines

Masterarbeit von

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Contents

1 Introduction **1**

1.1 Motivation 1

1.2 Problem Definition 1

1.3 Related Work 2

1.4 Our Contributions 2

2 Related Work **4**

2.1 Foundational paradigms of MLC 4

2.1.1 Binary Relevance 4

2.1.2 Label Powerset 5

2.1.3 Classifier Chains 5

2.2 Hierarchy aware architectures 6

2.2.1 From flat to hierarchical models 6

2.2.2 Local vs global approaches 6

2.2.3 Local approaches (Top down) 7

2.2.4 Global approaches (single classifier) 7

2.2.5 Encoding the Hierarchy with Transformers and Graph Neural Networks 8

2.3 The rise of Large Language Models 9

2.3.1 Zero shot and few-shot classification 9

2.3.2 LLMs for data augmentation 10

2.3.3 LLMs as evaluation assistants 10

2.4 Focus on data-specific challenges 11

2.4.1 Class imbalance and tail labels 11

2.4.2 Strategies for low resource and few-shot scenarios 12

2.4.3 Parameter-efficient fine-tuning for low-resource adaptation 12

2.4.4 Data augmentation with LLMs 13

3 Task and Data Analysis **15**
3.1 Task description 15
3.1.1 SemEval2025 Task 10: Overview 15
3.1.2 Task decomposition 15
3.1.3 Data and annotation 16
3.1.4 Participation and headline results 16
3.1.5 Evaluation of Subtask 2: Narrative Classification 17
3.2 Dataset Deep Dive 18

4 System architectures and methods **22**
4.1 Architecture 0: Transformer Baseline 22
4.1.1 Core architecture and model choice 22
4.1.2 Mitigating the class imbalance 23
4.1.3 Hierarchical-Conditioned training and data handling 24
4.1.4 Training Methodology 24
4.1.5 Addressing data scarcity and class imbalance 26
4.2 Architecture 1: Zero-Shot Autogen Framework 28
4.2.1 Motivation and core concepts 28
4.2.2 System components and architecture 29
4.3 Architecture 2: The Traceable Actor-Critique Pipeline 33
4.3.1 Motivation: From Decomposed Agents to Holistic, Reasoned Analysis 33
4.3.2 Pipeline architecture 34
4.3.3 Sequential workflow and processing stages 34
4.3.4 Key advantages of the evidence-grounded pipeline 37

5 Experimental setups **39**
5.1 Experimental setup 39
5.1.1 Hierarchical deBERTa baseline 39
5.1.2 Zero-Shot Agentic framework 40
5.1.3 Computational Environment 40
5.1.4 Actor-Critic pipeline 40

6 Experiments and results **41**
6.1 Synthesis of the multi-paradigm approach 41
6.2 Results for Architecture 0 (Transformer Baseline) 41
6.2.1 Experimental setup and evaluation metrics 42

6.2.2 Overall performance analysis 42
6.2.3 Language-specific performance analysis 45
6.2.4 Detailed performance analysis at optimal threshold 46
6.2.5 Key insights and limitations 48
6.3 Results for Architecture 1 (Zero-Shot Agentic Framework) 49
6.3.1 Overall Performance on English Test Set 49
6.3.2 Multilingual Performance Analysis 50
6.3.3 Critical implications for transformer baseline methodology 50
6.3.4 Detailed Error Analysis (on English DEV Set) 51
6.3.5 Key Insights and Limitations 52
6.4 Results for Architecture 2 (Actor-Critique Pipeline) 53
6.4.1 Experimental setup and evaluation framework 53
6.4.2 Ablation study: Impact of validation mechanism 54
6.4.3 Overall multilingual performance analysis 54
6.4.4 Key findings from comprehensive ablation study 54
6.4.5 Hierarchical performance characteristics 56
6.4.6 Language-specific narrative performance analysis 57
6.4.7 Subnarrative classification analysis 58
6.4.8 Comparative analysis with baseline architectures 60
6.4.9 Key insights and architectural implications 60
6.4.10 Cross-architectural comparison: AutoGen vs. Actor-Critique 62
6.4.11 Evaluation limitations and dataset size considerations 63
6.4.12 SemEval-2025 Task 10 competition performance 64

7 Conclusion **66**
7.1 Principal Findings and Implications 66
7.2 Key Open Challenges 67
7.3 Concluding Remarks 68

Bibliography **69**

Eidesstattliche Erklärung **77**

Abstract

The task of assigning multiple, hierarchically-structured labels to text documents, known as Hierarchical Multi-Label Classification (HMLC), is critical in domains from scientific archiving to legal analysis. This review traces the methodological evolution of HMLC, beginning with foundational problem transformation methods like Binary Relevance and Classifier Chains, which primarily address the challenge of label correlation. We then examine the paradigm shift introduced by pre-trained Transformers, dissecting the dichotomy between local, top-down approaches prone to error propagation and global, hierarchy-aware models that integrate structural constraints via specialized loss functions or Graph Neural Networks. Finally, we explore the current frontier, where Large Language Models (LLMs) are reframing the task through generative paradigms, enabled by techniques such as LLM-powered data augmentation, instruction fine-tuning, and Parameter-Efficient Fine-Tuning (PEFT). This narrative highlights a progression towards increasingly sophisticated methods for embedding hierarchical prior knowledge into statistical models.

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1 Introduction

1.1 Motivation

Text Classification (TC) is a foundational task in Natural Language Processing (NLP) [Zan+24]. While early work often framed TC as a multiclass problem, modern texts regularly contain multiple overlapping themes [Hu+25; TS18]. This motivates Multi-Label Classification (MLC), where instances can be assigned multiple labels simultaneously [TS18]. Beyond mere multiplicity, labels are frequently correlated [TS18; Hua+24], so modeling inter-label dependencies is crucial.

A particularly compelling setting is hierarchical multi-label classification (HMLC), where labels are arranged in a predefined hierarchy (e.g., tree or DAG) from coarse to fine categories [Liu+23]. This structure matches how humans reason about complex discourse. In the high-stakes context of online news and crisis communication, identifying nested narratives is both socially relevant and technically challenging. As a further motivation, this work aims to rigorously evaluate the classification capabilities of large language models (LLMs) on a difficult, nuanced HMLC task—one that requires fine-grained distinctions, hierarchical consistency, and evidence-grounded reasoning rather than shallow keyword matching.

1.2 Problem Definition

We ground our study in SemEval-2025 Task 10, Subtask 2 [Pis+25], a hierarchical narrative detection task for news. The objective is to assign each document multiple labels from a two-level taxonomy of narratives and subnarratives, i.e., mapping a text to a set of (narrative, subnarrative) pairs. The task features: (i) extreme class imbalance; (ii) hierarchical consistency constraints; and (iii) multilingual complexities. Formally, the MLC problem seeks to construct a function $f : X \to 2^L$ that maps an instance $x \in X$ to a subset $Y \subseteq L$, where L is the finite label set [TS18]. This function may be learned through supervised training or constructed via other means (e.g., zero-shot prompting). HMLC adds the requirement that predicted labels respect the hierarchical structure [Liu+23].

1.3 Related Work

Research in multi-label classification has progressed from early problem-transformation methods (Binary Relevance, Classifier Chains, Label Powerset) [Zha+18b; Wen+20; Sha+18] to Transformer-based models that learn contextual correlations [Dev+19] and then to hierarchy-aware architectures incorporating structure through specialized losses or graph formulations [SC22; Pen+21; Gon+20; Liu+23]. Yet in settings that are simultaneously long-tailed, multilingual, and hierarchically constrained, supervised approaches remain brittle: they overfit head labels, struggle with rare-node calibration, propagate errors top–down, and provide limited transparency about individual decisions. Recent zero-shot and agentic LLM approaches [SSB25b; SSB25a] reduce dependence on labeled data, but introduce new gaps in traceability, cost efficiency, and consistent enforcement of hierarchical constraints (true-path validity, evidence grounding). This trajectory motivates our contributions: systematically contrasting a strong supervised baseline with two zero-shot architectural paradigms while explicitly targeting the unresolved challenges of (i) long-tail robustness, (ii) native multilingual operation without translation-induced semantic loss, and (iii) auditable, evidence-grounded reasoning.

1.4 Our Contributions

We make the following contributions in the context of HMLC for narrative detection:

- We introduce a zero-shot agentic framework that decomposes narrative detection into specialized binary decisions handled by LLM agents, achieving competitive performance without task-specific fine-tuning.
- We propose a configurable, traceable actor - critic pipeline that augments the agentic approach with explicit evidence extraction and validation, improving robustness and interpretability.

- We provide a comparative analysis of three paradigms: a fine-tuned Transformer baseline, the agentic framework, and the actor - critic pipeline, highlighting trade-offs in accuracy, traceability, and cost.
- We conduct an in-depth error analysis on SemEval-2025 data, elucidating the roles of class imbalance, semantic ambiguity, and hierarchical constraints in model failures.

Paper outline. The remainder of this paper presents task and data analysis, details the system architectures, reports experiments and ablations, and concludes with discussion and future directions.

2 Related Work

2.1 Foundational paradigms of MLC

The main challenge in multi-label classification (MLC) was how to adapt algorithms designed for single-label (binary or multiclass) problems to handle multiple labels per instance. The core issue these methods faced was the presence of label correlations: in real-world data, labels are often not independent. For example, a news article tagged "Politics" is much more likely to also be tagged "Elections" than "Sports." Treating each label as a separate binary problem ignores these dependencies, potentially leading to suboptimal predictions.

Classical MLC methods can be seen as different strategies for balancing computational simplicity with the need to model label correlations. Some methods, like Binary Relevance, treat each label independently for simplicity, but this can miss important relationships between labels. Others, such as Classifier Chains or Label Powerset, explicitly model these correlations, but at the cost of increased computational complexity. The choice of method reflects a trade-off between efficiency and the ability to capture the true structure of the data.

2.1.1 Binary Relevance

Binary Relevance (BR) is the most intuitive approach. It decomposes the MLC problem with a label set of size $|\mathcal{L}|$ into independent binary classification problems. For each label, a separate classifier is trained to predict its presence or absence, effectively ignoring all other labels. [Zha+18b] The primary advantage of BR is its simplicity and efficiency, as the classifiers can be trained in parallel. Its main drawback, however, is the foundational label independence assumption, which completely disregards label correlations and can lead to lower predictive accuracy and logically incoherent label set predictions. [Suc+14]

2.1.2 Label Powerset

In contrast to decompositional methods, the Label Powerset (LP) method reframes the entire problem at once. It converts the multi-label task into a standard multi-class problem by mapping each distinct set of co-occurring labels to a single, unique class. For instance, the label sets 'Politics, Elections' and 'Sports, Weather' would become two separate classes for a multi-class classifier to learn. [Rea+11]

The main advantage of this approach is its ability to perfectly model the dependencies between labels for all combinations it has seen, as these correlations are baked into the class definitions [Suc+14]. However, this strategy is often impractical. The number of potential classes can become unmanageably large as the label set grows, a problem known as combinatorial explosion. This leads to a highly sparse class distribution where many label sets appear only a few times, making it difficult to train a robust model [CMM11]. Critically, the LP method cannot generalize to predict any combination of labels that did not appear in the training data.

2.1.3 Classifier Chains

The Classifier Chains (CC) method was proposed as a novel approach to overcome the stark trade-off between the label-independent BR and the computationally explosive LP. It seeks to model label dependencies while maintaining the efficiency of a binary relevance framework [Rea+11].

Like BR, CC trains $|\mathcal{L}|$ binary classifiers. However, instead of being independent, these classifiers are linked in a chain. The first classifier in the chain, $\mathcal{C}_1$, predicts the presence or absence of the first label. The predictions of this classifier are then used as additional features for the second classifier, $\mathcal{C}_2$, which predicts the second label. This process continues down the chain, with each classifier potentially benefiting from the predictions of all previous classifiers. This allows CC to capture label dependencies while still being relatively efficient to train. [Rea+11] The order of the chain can significantly impact performance, as earlier classifiers influence later ones [Rea+21]. To mitigate this, ensemble methods that average predictions over multiple random chain orders are often employed [Suc+14; Zha+18a].

2.2 Hierarchy aware architectures

The limitations inherent in classical problem transformation methods, particularly their struggles with large label spaces and their inability to deeply integrate structural information, precipitated a paradigm shift toward deep learning. We will focus on how neural architectures evolved from treating the label hierarchy as a post-hoc constraint to using it as a central component of the learning process. This evolution is characterized by a fundamental architectural divergence between local and global approaches, a conflict that was ultimately resolved through the powerful synthesis of Transformer-based text encoders and Graph Neural Networks (GNNs) for structure encoding.

2.2.1 From flat to hierarchical models

Early deep-learning methods for hierarchical multi-label classification often treated the task as a standard (flat) multi-label problem and ignored the label taxonomy. While these models learned strong text representations, they did not use the hierarchy's relationships. That matters because mistakes at higher levels of the taxonomy are semantically worse than small, nearby errors: for example, assigning a paper on "Quantum Mechanics" to "Arts" is much more serious than confusing "Physics" with "Chemistry." Flat models cannot distinguish these degrees of error. [Xu+21]

The paradigm shift occurred with the recognition that the hierarchy could be treated as a feature to guide learning, rather than just an output format. By making models "hierarchy-aware," it becomes possible to share statistical strength between parent and child nodes. For example, the few training instances available for a rare, specific label like "Superstring Theory" can be supplemented by the more abundant data from its parent labels, "String Theory" and "Theoretical Physics." This is especially crucial for improving performance on the long tail of infrequent labels that characterizes most real-world HMLC datasets. [Zan+24]

2.2.2 Local vs global approaches

The first generation of truly hierarchical models diverged into two main architectural philosophies: local and global. This division reflects a fundamental trade-off between capturing fine-grained, localized class relationships and maintaining a holistic, computationally tractable view of the entire label space.

2.2.3 Local approaches (Top down)

The local approach decomposes the hierarchical classification problem into a set of smaller, more manageable classification tasks distributed across the taxonomy. This is typically implemented as a top-down or "divide and conquer" strategy.

During inference, an instance is typically classified in a top-down manner. It is first evaluated by the classifier at the root; if a positive prediction is made for a node, the instance is then passed down to the classifiers of its children, and this process continues until a leaf node is reached or no further positive predictions are made. [RFR22]. While this approach excels at capturing the specific features that distinguish between closely related sibling classes, it suffers from a critical problem: **error propagation.** A single misclassification at a higher level of the hierarchy can irreversibly steer the prediction down an incorrect path, making it impossible to classify the instance into its correct, more specific sub-categories. [WCB18]

2.2.4 Global approaches (single classifier)

In contrast, the global approach uses a single, unified model to predict all labels in the hierarchy simultaneously. This is typically framed as a large multi-label classification problem where the output layer corresponds to the entire set of labels in the taxonomy. [WCB18]

The primary advantage of the global approach is that it inherently avoids the error propagation problem of local models, as all decisions are made in parallel by a single classifier. This makes the model more robust to errors at higher levels. Furthermore, global models are often more computationally efficient, as they require training only one model instead of a potentially large cascade of local classifiers. However, early global models faced a significant challenge: they struggled to effectively incorporate the complex structural information of the entire hierarchy into a single model. By treating the problem as a flat multi-label task, they often failed to capture the nuanced, local distinctions between sibling classes and could underfit the hierarchical relationships, thereby losing the very information that hierarchical classification aims to exploit. [WCB18; Zho+20]

2.2.5 Encoding the Hierarchy with Transformers and Graph Neural Networks

The solution to the local-versus-global problem came from combining two methods: Transformer models to understand text, and Graph Neural Networks (GNNs) to understand structure. This mix made it possible to build global models that are both efficient and aware of hierarchies. [WDH24; Zho+20]

GNNs are especially useful for working with label hierarchies. [Li+21] We can represent the taxonomy as a graph, where labels are nodes and parent-child links are edges. A GNN learns label representations by passing messages between connected nodes. Each node gathers information from its parents, children, and siblings. This way, the final label embeddings capture not just their own meaning, but also their place and relationships within the whole taxonomy.

A seminal work in this domain is the Hierarchy-Aware Global Model (HiAGM) by Zhou et al. (2020). HiAGM provides a blueprint for this new class of models. Its architecture consists of two main components: [Zho+20]

- A Text Encoder (e.g., BERT, RoBERTa) that generates a powerful contextualized representation of the input document.
- A Hierarchy-Aware Structure Encoder (e.g., a Bidirectional Tree-LSTM or a specialized Hierarchy-GCN) that operates on the label graph to produce hierarchy-aware label embeddings. This encoder models dependencies in both a top-down and bottom-up fashion, allowing information to flow in both directions across the hierarchy.

HiAGM further proposes two distinct fusion variants to combine text and structure features depending on the inference regime and efficiency/inductivity trade-offs.

HiAGM-LA (Multi-Label Attention) An inductive approach that uses a multi-label attention mechanism to compute document-specific label representations by attending from the text encoder outputs to the hierarchy-aware label embeddings. Because the attention weights are computed per document, the model can generalize to unseen instances without explicitly re-running a graph propagation step for each input, label embeddings can be pre-computed and stored, and the attention operation is applied at inference time.

HiAGM-TP (Text Feature Propagation) A deductive approach that propagates text features across the label hierarchy via graph propagation: document features are attached to label nodes and a GNN is run to diffuse these features through the taxonomy. This requires executing the GNN at inference time for each instance (or batch), which can be more computationally expensive but allows richer instance-specific propagation of evidence through the hierarchy.

The principles demonstrated by HiAGM have been extended and refined in subsequent work. For instance, Xu et al. (2021) proposed a framework using a loosely coupled GCN to explicitly model not only the vertical correlations (parent-child dependencies) but also the horizontal correlations (relationships between sibling nodes at the same level). This allows the model to capture, for example, that "Computer Science" and "Electrical Engineering" are more closely related to each other than to "History," even if they share the same parent ‘Science & Technology’.

2.3 The rise of Large Language Models

The advent of Large Language Models (LLMs) has initiated another profound shift in the NLP landscape. Their advanced reasoning and generation capabilities are redefining their role in the HMLC pipeline, moving them beyond being simple end-point classifiers to becoming integral co-pilots in various stages of the workflow.

2.3.1 Zero shot and few-shot classification

The most immediate impact of LLMs is their remarkable ability to perform classification tasks with little to no task-specific training data. Through in-context learning, an LLM can be prompted with a task description and a few examples (few-shot) or even no examples (zero-shot) and perform hierarchical multi-label classification. This capability represents a paradigm shift for low-resource scenarios, potentially obviating the need for extensive data collection and annotation efforts; for narrative detection, one can provide definitions of the target narratives and ask the LLM to classify a new document accordingly. [Wan+23]

For instance, Eljadiri and Nurbakova (2025) demonstrate a high-performing zero-shot agentic approach to narrative classification. Their system instantiates multiple specialized LLM agents, each responsible for a binary decision on a single narrative or subnarrative label, while a meta-agent aggregates the binary outputs into final multi-label predictions. The agents are orchestrated with AutoGen and operate without task-specific fine-tuning, which enables parallel detection across the two-level taxonomy; this design yielded competitive results on the SemEval-2025 test set and highlights the practical zero-shot LLM systems for complex hierarchical tasks. [EN25]

2.3.2 LLMs for data augmentation

One of the most promising applications of LLMs is in addressing data scarcity and class imbalance through synthetic data generation. An LLM can be prompted to act as a domain expert and generate new text samples for underrepresented (tail) classes. This can be done by paraphrasing existing examples to increase diversity or by generating entirely new, plausible examples from scratch. This approach offers a sophisticated and flexible alternative to traditional data augmentation techniques like synonym replacement or back-translation. [CSB25; GZ24]

In more extreme cases; for example, a scenario with complete lack of labeled data, an LLM can be used to generate a large corpus of "weak" or "silver" labels. These labels, while imperfect, can serve as a starting point for training a smaller, more efficient model. Frameworks like JSDRV by Yang et al. (2024) take this a step further, using a reinforcement learning policy to select the highest-quality LLM-generated annotations for fine-tuning, creating a self-improving annotation and training pipeline [Yan+24].

2.3.3 LLMs as evaluation assistants

For extreme multi-label classification tasks with thousands of labels, manual evaluation is prohibitively expensive and time-consuming. Li et al. (2023) aimed to compare a label-ranking BiCross-Encoder against a SciBERT classifier in a very large-label setting (11,486 labels) while lacking a ready human-annotated test set. They demonstrated a practical approach by asking ChatGPT to rate the relevance of candidate labels: for each document they fed the top-10 model-predicted labels to ChatGPT and requested a 3-point relevance score (0 = irrelevant, 1 = somewhat relevant, 2 = highly relevant) along with brief explanations. The authors treated these LLM judgments as automatic annotations to compare models, then spot-checked a sample with subject-matter experts and reported about 60% agreement on the 3-point scale (rising to 82% when collapsing 1 and 2 into a single “relevant” class). They conclude that using

ChatGPT as an evaluation assistant is a cost- and time-efficient way to bootstrap evaluation when comprehensive human annotation is infeasible. [Li+23]

2.4 Focus on data-specific challenges

While architectural innovations have driven significant progress, the performance of any HMLC model is ultimately constrained by the quality and characteristics of the available data. Real-world datasets for narrative detection and other complex classification tasks are rarely pristine; they are often imbalanced, small, and increasingly, multilingual.

2.4.1 Class imbalance and tail labels

Perhaps the most pervasive challenge in real-world classification is extreme class imbalance, often described as a long-tailed distribution. In such datasets, a small number of "head" classes are represented by a vast number of training examples, while the majority of "tail" classes are represented by very few, sometimes single-digit, examples. [Hua+21] This constitutes a severe challenge for standard training algorithms, which, when optimized for overall accuracy, tend to become biased towards the majority classes and perform poorly on the infrequent but often more interesting tail classes.

The inadequation of conventional resampling

In single-label classification, a common strategy to combat imbalance is data resampling, either oversampling the minority classes or undersampling the majority classes [Luq+19; Tha+20]. However, this approach is fundamentally ill-suited for the multi-label context. An instance in an MLC dataset can simultaneously belong to multiple classes, for example, one highly frequent head class and one extremely rare tail class. If one were to oversample this instance to increase the representation of the tail class, one would inadvertently also increase the representation of the already-dominant head class, potentially exacerbating the overall imbalance with respect to other labels. This entanglement makes simple resampling ineffective and often counterproductive. [Yua+24]

A particularly effective technique that has been instancied is the distribution-based loss function. As demonstrated by Huang et al. (2021), this approach goes beyond simple re-weighting by class frequency. It introduces a loss formulation that inherently addresses both the class imbalance and the label linkage (co-occurrence) problems simultaneously. The re-weighting scheme is designed to account for the fact that an instance with multiple labels can be oversampled if each of its labels is treated independently [Hua+21].

2.4.2 Strategies for low resource and few-shot scenarios

When unlabeled data is abundant but labeling is expensive, active learning provides a principled way to build a high-quality training set with minimal human effort [Li+23]. Instead of randomly selecting data for annotation, an active learning system iteratively queries a human oracle (e.g., a domain expert) to label the instances that the model is most uncertain about or that are expected to provide the most information for improving the model. For XMC, Li et al. (2023) propose a sophisticated active learning pipeline that uses a Bi-Encoder model to first retrieve a manageable pool of potentially relevant documents for new, unlabeled classes. A greedy acquisition strategy then selects the most promising candidates from this pool for expert annotation, drastically reducing the labeling burden compared to annotating the entire corpus.[Li+24]

2.4.3 Parameter-efficient fine-tuning for low-resource adaptation

Fully fine-tuning a large PLM with billions of parameters on a small dataset is not only computationally expensive but also prone to overfitting. Parameter-Efficient Fine-Tuning (PEFT) methods have emerged as a critical technology for low-resource adaptation. [Su+24] Techniques like Adapters and Low-Rank Adaptation (LoRA) work by freezing the vast majority of the pre-trained model's weights and introducing a very small number of new, trainable parameters. LoRA, for example, injects trainable low-rank matrices into the layers of the Transformer, allowing the model to adapt to the new task by learning low-rank updates to its original weight matrices. By drastically reducing the number of trainable parameters (often by over 99%), PEFT makes the fine-tuning process more stable on small datasets, reduces memory requirements, and helps prevent catastrophic forgetting of the knowledge learned during pre-training. [Liu+24; Raz+24]

2.4.4 Data augmentation with LLMs

As discussed in Section 2.3.2, LLMs can serve as powerful data generators. For low-resource classes, an LLM can be prompted to create synthetic training examples by paraphrasing the few existing seed examples or by generating novel instances based on a class description. A cost-benefit analysis by Cegin et al. (2025) reveals that this approach provides the most significant advantage when the number of initial seed examples is extremely small (e.g., one-shot or few-shot learning) [CSB25]. As the amount of available data increases, the performance gains from LLM augmentation may diminish relative to its computational cost, and in some cases, traditional augmentation techniques can achieve comparable results more efficiently. [CSB25]

Concrete evidence of these trade-offs is provided by Whitehouse et al. (2023) [WCA23], who systematically investigate LLM-powered augmentation for multilingual commonsense reasoning. The authors use a variety of LLMs (Dollyv2, StableVicuna, ChatGPT and GPT4) to generate synthetic training examples for three low-resource benchmarks (XCOPA, XWinograd and XStoryCloze). They compare generation strategies, producing data directly in the target language, generating in English and translating the output, and using English-generated data as-is, and then fine-tune compact multilingual models (mBERT and XLMRoBERTa) on the augmented corpora.

Their results show that augmenting scarce training sets with high-quality LLM generations can substantially improve downstream accuracy (up to a 13.4point increase in the best case). GPT4 and ChatGPT produce the most natural and logically coherent examples in most languages, with GPT4 often yielding the largest gains; smaller/open models (Dollyv2, StableVicuna) deliver weaker, less consistent improvements. The authors also perform a human evaluation: native speakers rate the naturalness and logical coherence of generated examples, confirming GPT-4's superior quality but revealing languagedependent failures (notably for some lowresource languages such as Tamil). Overall, the study demonstrates that LLMs are a practical and powerful source of synthetic data for crosslingual commonsense tasks when reliable generation is available, while also highlighting limitations in coverage and the need for quality control.

In summary, the field has progressed from classical HMLC methods to sophisticated hierarchy-aware deep learning models. While recent work in the SemEval-2025 shared task has highlighted the promise of LLM-based prompting, these approaches often rely on monolithic, multi-step prompts. This paper explores a different paradigm: decomposing the HMLC task into a system of independent, specialized agents. We investigate both a parallel agentic framework and a novel actor-critic pipeline designed for enhanced traceability, directly addressing the need for more modular and reliable systems in zero-shot narrative classification.

3 Task and Data Analysis

3.1 Task description

3.1.1 SemEval2025 Task 10: Overview

SemEval-2025 Task 10 introduced a multilingual benchmark for the analysis of news narratives and entity framing across five languages and two high-impact domains (Ukraine–Russia War, Climate Change). The task is decomposed into entity framing, narrative classification, and narrative extraction subtasks, combining multi-label classification and evidence-grounded generation. Participating teams demonstrated the effectiveness of hierarchical prompting and targeted context selection; however, grounded explanation generation and label imbalance across languages remain unsolved challenges [PM25; Fan+25; SSB25a; F25].

We participated in this shared task (together with Diana Nurbakova) as competitors in Subtask 2 (Narrative Classification); the experiments reported here were carried out in that shared-task context.

This shared task asked teams to build systems that identify and extract dominant narratives from multilingual news, prioritising evidence-grounded explanations and robustness across languages. We ran multilingual evaluations to compare per-language behaviour.

3.1.2 Task decomposition

The shared task was split into three subtasks:

1. **Entity Framing (Subtask 1)**: Given an article and named-entity mentions, assign main and fine-grained role labels to each mention (multi-label, multi-class role classification). Grounding to spans and role taxonomies was required. [PM25]

2. **Narrative Classification (Subtask 2)**: Assign articles one or more narrative labels from a two-level taxonomy (main narrative → subnarrative). This is a multilingual multi-label document classification problem. [PM25]

3. **Narrative Extraction (Subtask 3)**: For a given article and its dominant narrative, generate a concise (max. 80 words) free-text explanation supporting that narrative, grounded in evidence from the article. This subtask focused on faithfulness and coverage of extracted explanations. [PM25]

3.1.3 Data and annotation

Training and evaluation data were collected from online news sources spanning 2022-mid-2024. The dataset includes articles from a mix of mainstream and lower-credibility outlets; annotations follow multilingual guidelines and predefined taxonomies for entity roles and narratives. [Sem25b]

3.1.4 Participation and headline results

The task attracted broad participation; hundreds of teams registered and dozens submitted official test results. Representative high-performing approaches included:

- **Fane et al. ("Fane")** - zero-shot LLM prompting with a hierarchical decomposition for entity framing; reported Main Role Accuracy ≈ 89.4% and Exact Match Ratio ≈ 34.5% on English. [Fan+25]

- **GateNLP** - *Hierarchical Three-Step Prompting (H3Prompt)* for Subtask 2 (narrative classification): cascade of domain → main narrative → subnarrative prompting; top performer on the English test set among competing teams. Code published. [SSB25a]

- **DEMON / TechSSN / LTG / BERTastic** - representative competitive systems using fine-tuning of large pre-trained encoders (e.g., XLM-R / DeBERTa variants) or fine-tuned LLaMA-family models for framing and narrative tasks. Several system descriptions and official proceedings papers document model design choices and per-language performance differences. [F25; P25; RN25; Nak+25]

These system papers illustrate a common trend: hierarchical or multi-step prompting and careful context selection (to fit model context windows) were effective strategies across teams. [Fan+25; SSB25a; RN25]

3.1.5 Evaluation of Subtask 2: Narrative Classification

The second subtask of SemEval-2025 Task 10 required participants to assign *narrative* (coarse-level) and *sub-narrative* (fine-level) labels to each news article. Since articles can be associated with multiple narratives, this constitutes a multi-label, multi-class document classification problem [PM25].

Evaluation target. Participants had to provide two sets of predictions per article: (1) a set of coarse-level narrative labels and (2) a set of fine-grained sub-narrative labels. Although both levels were evaluated, the official ranking was based on performance at the sub-narrative level, as this captures more nuanced framing [Sem25b].

Metrics. Two main evaluation metrics were applied:

- **Macro F1 (coarse level).** This is the unweighted mean of F1-scores computed for each coarse narrative class. It penalizes models that ignore low-frequency labels.
- **F1 (samples).** This metric is computed at the level of individual articles. For each document, precision, recall, and F1 are compute by comparing the predicted label set with the gold label set. The final score is the average of these values across all documents.

In addition, standard deviations across multiple runs and across languages were reported on the leaderboard to quantify robustness [Sem25a].

Worked example. Consider a simplified case with three possible sub-narratives: $\{N_1, N_2, N_3\}$. Suppose the gold labels for a document are $\{N_1, N_3\}$, while a system predicts $\{N_1, N_2\}$.

- **Precision** $= \frac{|\{N_1,N_2\}\cap\{N_1,N_3\}|}{|\{N_1,N_2\}|} = \frac{1}{2} = 0.5.$
- **Recall** $= \frac{|\{N_1,N_2\}\cap\{N_1,N_3\}|}{|\{N_1,N_3\}|} = \frac{1}{2} = 0.5.$
- **F1 (sample)** $= \frac{2\times0.5\times0.5}{0.5+0.5} = 0.5.$

In contrast, if the system had predicted only $\{N_1\}$, precision would be 1.0, recall 0.5, and F1 (sample) 0.67. This illustrates that partial but precise predictions may yield a higher F1 than predicting extra incorrect labels.

Implications. Because the leaderboard ranks systems by fine-level labels using sample-based F1, models must balance recall (capturing all relevant sub-narratives) with precision (avoiding irrelevant ones). Macro F1 further incentivizes attention to rare narratives, since underperformance on them disproportionately lowers the score. [PM25].

3.2 Dataset Deep Dive

We conducted a systematic descriptive analysis of the released training split to quantify its distributional, linguistic, and structural properties. Our analysis reveals several key characteristics regarding its scale, multi-label nature, class imbalance, and co-occurrence structure, which collectively motivate our architectural design choices.

Scale and Multilinguality. The corpus contains 1,699 documents in five languages: Bulgarian (BG), Portuguese (PT), English (EN), Hindi (HI), and Russian (RU). The distribution is relatively balanced across Bulgarian, Portuguese, and English ($\approx$ 399–401 documents each), with 366 Hindi documents and 133 Russian documents. This multilingual context is critical for assessing the cross-lingual generalization capabilities of classification models.

Multi-Label Nature. Building upon this multilingual foundation, a defining feature of the dataset is the high frequency of multi-label annotations. A majority of documents (54.0%) are assigned more than one narrative, with an average of 2.28 narrative labels per document and a maximum of 8. This observation confirms that multi-label modeling is essential; treating the task as a series of independent single-label classifications would discard significant structural information present in the data.

Label Frequency and Extreme Imbalance. The complexity of the multi-label structure is further compounded by severe class imbalance. The taxonomy consists of 22 distinct narrative labels and 95 sub-narrative labels, distributed in a strongly skewed manner. The most frequent narrative appears 584 times (representing $\approx 15.1\%$ of all narrative annotations in the multilingual training set), while the least frequent occurs only 9 times. This yields an imbalance ratio of approximately 65:1, creating a classic long-tailed distribution. Notably, the dataset also includes a generic Other category for documents that do not fit any predefined narrative, which accounts for a substantial portion of the data and requires models to learn a negative class boundary against all defined narratives. The most prevalent narratives are dominated by war-related topics (e.g., 'URW: Discrediting Ukraine', 'URW: Discrediting the West, Diplomacy'), highlighting the dataset's focus on high-stakes domains. This sparsity poses a significant challenge for supervised models, which can become biased towards majority classes and fail to generalize to rare but meaningful narratives.

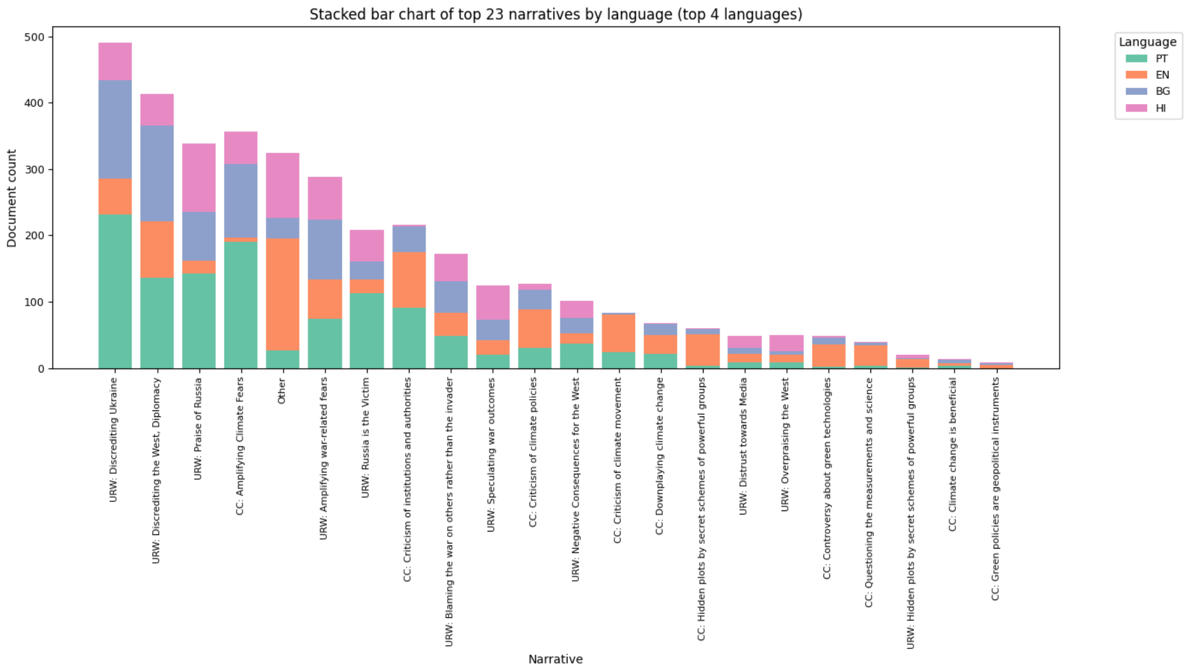


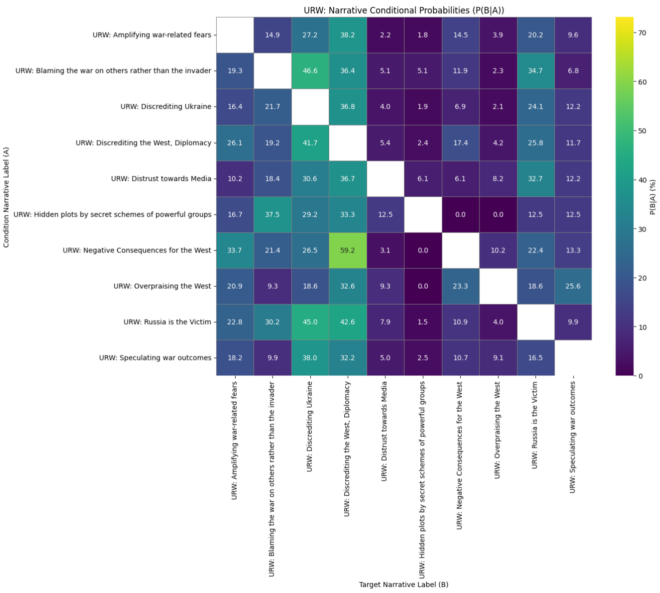
Figure 3.1: Long-tailed distribution of narrative labels in the SemEval-2025 training set. The bar chart shows the frequency of each coarse narrative, illustrating that a small number of narratives account for a large fraction of annotations while many narratives have very few examples. This severe imbalance motivates the use of zero-shot and evidence-grounded LLM approaches that do not rely on large per-class training counts.

Structured Label Co-occurrence. Despite this imbalance, the relationships between labels are not random but exhibit meaningful patterns. Co-occurrence analysis reveals that while most pairs of narratives rarely appear together, certain thematically-linked pairs exhibit strong associations. For example, documents often combine *attack* and *counter-attack* narratives (e.g., discrediting Ukraine together with discrediting the West), or pair attacks on an adversary with *praise of allies* (e.g., discrediting Ukraine alongside praise of Russia). Other recurrent patterns reflect causal framing, such as linking sanctions to economic hardship, or pairings between diplomatic failure and critiques of legitimacy. Military operations are frequently co-annotated with victim/threat framings, and conspiracy claims commonly appear together with explicit attribution to external actors. These recurring narrative couplings illustrate that co-occurrence captures structured discourse strategies in which labels reinforce one another. More advanced analysis further reveals directional dependencies (i.e., $P(B \mid A) eq P(A \mid B)$).

Language-Specific Patterns. These structural complexities are further nuanced by cross-lingual variations. The dataset exhibits notable variation across languages, with the average number of labels per document (label density) highest in Portuguese ($\approx$ 3.04) and lower in Hindi ($\approx$ 1.79), suggesting potential differences in source document complexity or annotation practices. Furthermore, the prevalence of specific narratives and their co-occurrence patterns differ across languages. This cross-lingual heterogeneity underscores the importance of using language-aware methods or, at a minimum, explicitly validating a model's performance on each language.

Document Length Heterogeneity. Finally, the dataset presents additional challenges through its variable document lengths. The corpus exhibits high variability, with a mean of approximately 404 words, a median of 371 words, and a range spanning from 45 to 4,422 words. This heterogeneity necessitates the use of effective truncation or long-text handling strategies in model design to manage computational constraints without losing critical context.

(a) CC narratives heatmap



(b) URW narratives conditional graph

Figure 3.2: Co-occurrence / conditional-probability visualizations for the most frequent narrative groups. Top: heatmap for CC narratives (normalized cell values). Bottom: conditional-probability visualization for URW narratives (directional). These paired views help identify both symmetric and directional dependencies among narratives.

4 System architectures and methods

To address the challenges of hierarchical multi-label classification presented by the SemEval-2025 dataset, we designed and evaluated three distinct architectural paradigms. We began by establishing a supervised baseline using a fine-tuned Transformer model. We then developed two zero-shot systems that leverage the advanced reasoning capabilities of Large Language Models (LLMs): a parallel agentic framework and a sequential, traceable actor-critic pipeline. This section details the design of each architecture.

4.1 Architecture 0: Transformer Baseline

To establish a robust supervised baseline, we developed a custom Transformer-based architecture explicitly designed to model the two-level taxonomy of the task. Instead of treating the label space as flat, our Multi-Head mDeBERTa for Hierarchical Classification directly encodes the parent-child relationships into its structure. This approach represents a state-of-the-art technique for Hierarchical Multi-Label Classification (HMLC), moving beyond soft regularization penalties to a more principled, hard-coded hierarchical design.

4.1.1 Core architecture and model choice

The foundation of our model is **mDeBERTaV2** (microsoft/mdeberta-v2-base) [He+21], a powerful Transformer encoder chosen for its architectural innovations that are particularly relevant to nuanced understanding tasks. While mDeBERTaV2 is a strong general-purpose model, its **disentangled attention mechanism** provides a compelling theoretical advantage for our specific problem.

Standard Transformer attention mechanisms [Vas+17] compute attention weights based on a single vector that combines both a token's content (what it is) and its relative position (where it is). In contrast, mDeBERTa's disentangled attention uses two separate vectors for each token: one representing its content and another representing its relative position. Attention weights are then computed based on the interactions between the content and position vectors of different tokens [He+21].

The intuition for our task is as follows: Narrative classification often depends on understanding the relationships between specific keywords (content) and their structural arrangement within a sentence or paragraph (position). For example, the narrative "URW: Blaming the war on others rather than the invader" relies on identifying entities like "the West" or "Ukraine" (content) and their syntactic roles as agents or recipients of blame (a position-dependent relationship). By disentangling content from position, mDeBERTa can more explicitly model these types of syntactic and relational dependencies, which are crucial for distinguishing between semantically similar but structurally different narratives.

While this mechanism offers a principled advantage, we also approached this choice experimentally. A key goal of implementing this advanced baseline was to test the limits of the fine-tuning paradigm: if even a model with a sophisticated, disentangled attention mechanism fails to learn from the sparse data, it provides stronger evidence that the problem lies with the learning paradigm itself, not just the specific model architecture.

Anticipated limitations. Based on our data analysis, we anticipated that a vanilla approach to finetuning would face significant limitations. The primary concern is its susceptibility to the severe class imbalance. By optimizing for overall accuracy, the standard BCE loss incentivizes the model to achieve high performance on the high-frequency "head" classes, often by simply predicting "negative" for the vast majority of rare "tail" classes. This behavior is expected to result in poor performance on metrics sensitive to class imbalance, such as the macro-averaged F1 score, and serves as the primary motivation for the more advanced baselines and zero-shot architectures explored subsequently.

4.1.2 Mitigating the class imbalance

To directly address the class imbalance issue, we enhance the loss function with positive class weighting. This technique modifies the standard BCE loss by increasing the penalty for misclassifying a positive instance of a rare class. For each class c we compute a per-class weight $\mathrm{pos\_weight}_c$ as the ratio of negative to positive training counts. Let N_c and P_c denote the number of negative and positive training examples for class c, respectively:

$$\mathrm{pos\_weight}_c = \frac{N_c}{P_c}.$$

This weight is then integrated directly into the BCE loss calculation, effectively increasing the cost of a false negative for rare classes. For a common label where positive and negative samples are balanced, the weight is close to 1, behaving like the standard BCE loss. However, for a rare label with a **pos_weight** of, for example, 65, the loss incurred for failing to predict that label when it is present is magnified 65 times.

In practice we pass the vector of per-class positive weights to the BCEWithLogitsLoss (or equivalent) implementation so rare classes contribute proportionally more to the loss. The approach is intended to improve recall on tail classes and, consequently, boost Macro F1 by forcing the model to pay closer attention to underrepresented labels.

4.1.3 Hierarchical-Conditioned training and data handling

4.1.4 Training Methodology

Our multi-head mDeBERTa architecture employs a training and data handling strategy designed to address the challenges of hierarchical structure, class imbalance, and data scarcity.

Hierarchically-Conditioned Training. The effectiveness of our multi-head mDeBERTa architecture is realized through a specialized training methodology designed to enforce the taxonomy's structure and combat the severe data imbalance. The total loss is a carefully calculated sum of losses from the parent-level classifier and the conditionally activated child-level heads.

Parent-Level Training. At the top level, the parent classifier is trained to predict the presence of all coarse-grained narratives. It is optimized using **Binary Cross-Entropy with Logits Loss** (BCEWithLogitsLoss). To counteract the significant class imbalance among parent narratives, we apply **positive class weighting**. For each parent label, a weight inversely proportional to its frequency is computed, compelling the model to assign a higher penalty for misclassifying rare parent narratives.

Child-Level Conditional Training. The core of our hierarchical approach lies in the conditional training of the child-level heads. This process ensures that each specialized child classifier is trained exclusively on relevant data. For a given training document, the loss for a specific child-classifier head (for example, the head for URW: Discrediting Ukraine's children) is calculated if and only if that document is labeled with the corresponding parent narrative. If the parent label is absent, the child head is "inactive" for that instance and contributes zero to the total loss. This conditional activation hard-codes the true-path rule into the training loop, preventing the child classifiers from being overwhelmed by the vast majority of documents where their parent narrative is not present.

A concrete example. Consider a document labeled with the parent narrative P_1 and its child sub-narrative C_{1a}. During the forward pass:

- The parent-level loss is calculated across all parent labels, with the model being rewarded for predicting P_1 as positive.
- The system checks the ground-truth labels. Since P_1 is present, the specific child-classifier head for P_1's children is activated.
- The loss for this child head is calculated, rewarding the model for predicting C_{1a} as positive.
- All other child-classifier heads (for P_2, P_3, etc.) remain inactive for this document.

This focused training strategy allows each child head to become a highly specialized expert, learning the subtle distinctions between its sibling sub-narratives without being distracted by irrelevant data from other branches of the hierarchy.

Figure 4.1 provides a schematic overview of the Multi-Head mDeBERTa architecture and the conditional activation behaviour described above.

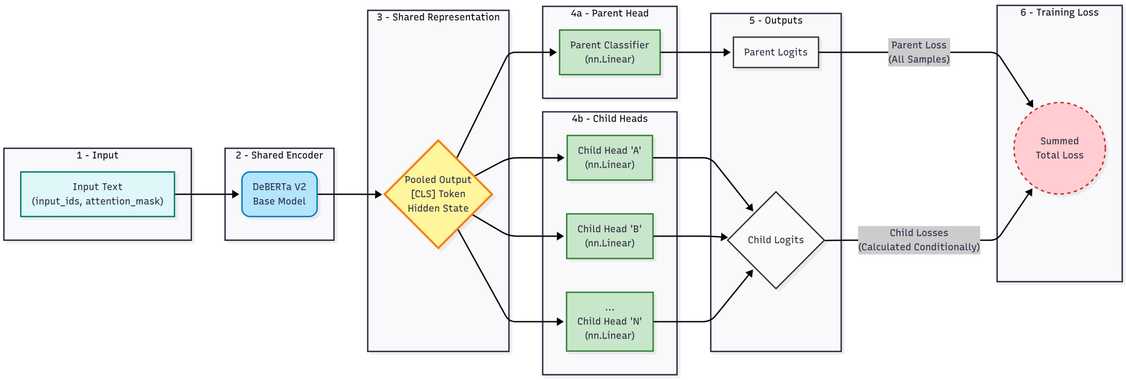


Figure 4.1: Multi-Head mDeBERTa architecture for hierarchical narrative classification. The shared encoder produces a pooled representation that is consumed by a parent-level classifier and multiple conditionally-activated child heads; training aggregates parent loss across all samples while child losses are computed conditionally.

4.1.5 Addressing data scarcity and class imbalance

Data pre-processing

Overview. Now, a critical first step in our methodology was a simple data pre-processing pipeline. As the data is already scarce, we cannot afford to have mDeBERTa learn from noisy inputs. The raw documents, sourced from online news articles, contained non-content artifacts typical of web scrapes, such as residual HTML tags, navigation links, and boilerplate phrases (e.g., "Share this article"). This noise both dilutes the core narrative content and consumes a valuable portion of the model's input context window (mDeBERTa uses up to 512 tokens).

Cleaning operations. The pipeline aimed to isolate the main body of each article and produce clean, normalized text. Typical operations included removing HTML and scripting remnants, filtering common boilerplate phrases, and normalizing whitespace to ensure consistent formatting. These steps raise the signal-to-noise ratio of the inputs and help models focus their representational capacity on semantically relevant content.

Implementation detail. Rather than constructing a suite of regular expressions or deterministic heuristics, we found **Qwen3-4b-Thinking-2507** to be a highly effective and low-effort option for this task. The LLM was used to identify and delete non-content artifacts while preserving the surrounding text verbatim; in practice, this approach removed navigation elements and calls-to-action embedded inside paragraphs while keeping the original article intact.

Multilingual data augmentation via translation

Motivation. The primary challenge in training the fine-tuning baseline is the severe scarcity of training examples, particularly for the tail-end classes. While the dataset is multilingual, the number of documents in any single language is modest. To significantly expand the training corpus and expose the model to more diverse linguistic expressions of the same narratives, we implemented a data augmentation strategy centered on high-fidelity translation.

Methodology. We leveraged the state-of-the-art translation capabilities of a large language model (Gemini 2.5 Flash) to augment our training data. Our process was as follows:

1. **Translation to a high-resource language**: For each document in the non-English training sets (Bulgarian, Portuguese, Hindi, and Russian), we generated a corresponding English version. We chose English as the target language due to the superior performance of modern LLMs in translating into English, which helps preserve nuanced meaning, tone, and authorial intent.
2. **Corpus expansion**: The newly translated English documents were then added to the training set alongside the original, untranslated documents. This procedure effectively doubles the size of our non-English training data; for example, the 399 Portuguese documents yielded an additional 399 English translations, both used for training.

Intended outcome. This augmentation strategy provides two key benefits. First, it dramatically increases the volume of training data, which helps mitigate overfitting for a large model like mDeBERTa. Second, by training on both the original and translated versions, the multilingual mDeBERTa model is encouraged to learn language-invariant representations of the narratives: it learns to associate a specific narrative with both its original linguistic expression (e.g., in Portuguese) and its English equivalent, pushing it to focus on the core semantic concepts rather than surface-level lexical features. The resulting classifier should be more robust and generalizable across languages.

Handling the "Other" Class

Motivation. The "Other" class represents a substantial portion of the dataset ($\approx 8\%$) and presents a unique modeling challenge. Unlike defined narrative labels, it signifies the *absence* of any specific narrative and, in the training data, never co-occurs with other labels. Including it directly in a multi-label framework can be problematic, as the model may struggle to learn a clear decision boundary for such a broad and negatively-defined category.

Methodology. To address this, we adopted a strategy of treating the "Other" class as an implicit negative case. We removed all instances labeled exclusively as "Other" from the training set. Consequently, the model was trained *only* on documents that contained at least one specific, defined narrative. During inference, this training strategy translates to a simple and effective decision rule: if the model, after applying a sigmoid activation and a confidence threshold, predicts **zero** positive labels for a given document, that document is then assigned the "Other" label.

Alternative (note).

> *We briefly considered treating "Other" as an explicit class and using the model's predicted probability for it as an adaptive threshold (i.e., a candidate label must exceed the "Other" probability to be assigned). We leave this variant for future work.*

Intended outcome. This approach simplifies the learning objective for the Transformer model. Instead of learning to positively identify the features of the "Other" class—a vague and potentially unbounded task—the model can focus entirely on learning the positive features of the defined narratives. This reframes the problem as learning to recognize "something" versus "nothing," a much clearer and more constrained task intended to improve the precision of the defined narrative predictions.

4.2 Architecture 1: Zero-Shot Autogen Framework

4.2.1 Motivation and core concepts

While the enhanced baseline models are designed to mitigate the challenges of class imbalance and hierarchical inconsistency, they are fundamentally limited by the scarcity of training data for many labels (we remind that the train set contains 22 unique narratives, while the taxonomy defines 23). To overcome this dependency on large, labeled datasets, we explored a fundamentally different paradigm that shifts from learning from examples to reasoning from instructions. This led to the development of our first contribution: a zero-shot agentic framework that leverages the pre-existing world knowledge and reasoning capabilities of Large Language Models (LLMs).

Inspiration Our design was inspired by the classical Binary Relevance (BR) method, which decomposes a multi-label problem into a set of independent binary classification tasks [Zha+18b]. BR's primary strength is its simplicity and modularity, but its critical weakness is the assumption of label independence. We sought to create a system that retained the modularity of BR while re-introducing a notion of label relationships in a structured, hierarchical manner. Furthermore, we were motivated to investigate the zero-shot classification capabilities of modern LLMs on this task.

To implement this vision, we used AutoGen [Wu+24], an open-source framework for multi-agent LLM conversations. AutoGen allows users to define multiple agents with specific roles and manages the conversation flow between them. In our case, we created one agent for each narrative label and used AutoGen to coordinate their interactions during the classification process.

The core concept of our framework, first introduced in [EN25], is to decompose the HMLC task into a two-stage, hierarchical query process handled by a multi-agent system.

4.2.2 System components and architecture

Agentic multi-agent framework

The agentic framework is constructed from a set of conversational agents, each instantiated as an LLM with a narrowly defined role and area of expertise. Our architecture comprises three distinct agent types:

- **User Proxy Agent.** The primary interface to the system. It receives a document for classification from an external user and initiates the workflow by passing the text to the Manager Agent. This is a AutoGen specific requirement to bootstrap the conversation.
- **Manager Agent (Orchestrator).** The central coordinator that directs the classification process. The Manager selects which agents to query at each stage, manages conversational group chats, and aggregates the final predictions into a structured multi-label output.

- **Narrative and Sub-narrative Agents (Specialists).** For every label in the two-level taxonomy we instantiate a dedicated specialist agent. Each such agent is initialized with a system prompt containing the label's official definition and examples, and its single task is to decide whether its assigned narrative is present in the document.

This multi-agent design enables strong specialization: each agent focuses its reasoning on a narrow, well-defined sub-problem and collectively contributes to the final, complex classification.

The *propose and refine* classification workflow

The classification of a document proceeds through a structured, multi-stage workflow that mimics a "propose-and-refine" strategy, as illustrated in Figure 4.2. Instead of a simple broadcast, the system uses Manager Agents at each level of the hierarchy to act as initial, coarse-grained classifiers, whose proposals are then validated by specialized agents.

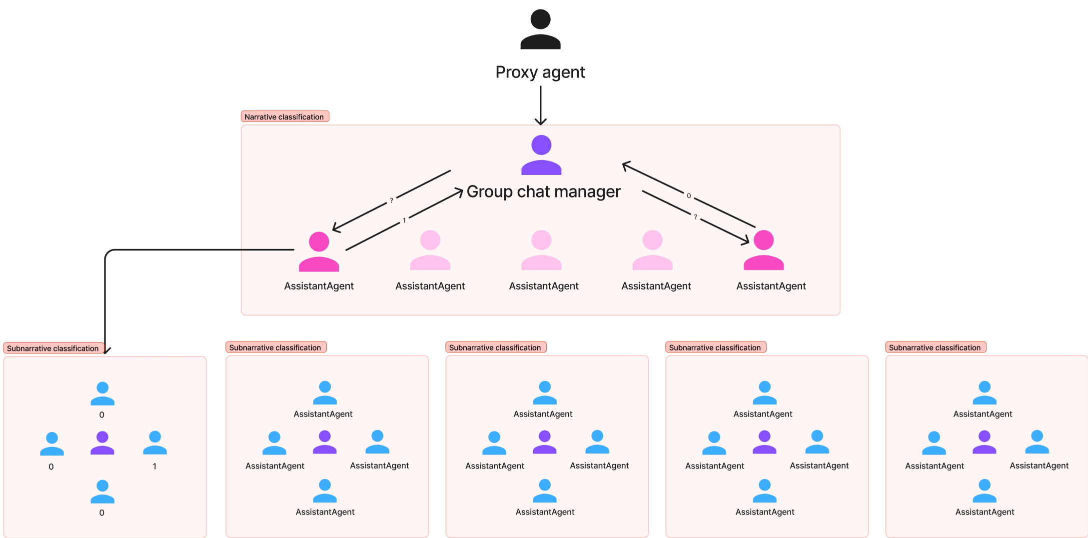


Figure 4.2: AutoGen multi-agent architecture showing the hierarchical propose-and-refine workflow. The Proxy Agent initiates the process, the Group Chat Manager coordinates narrative-level classification, and separate sub-managers handle sub-narrative classification for each confirmed parent narrative.

Stage 1: Narrative-Level Proposal and Refinement Upon receiving a document from the User Proxy, the top-level Manager Agent initiates the classification. It first performs an internal, high-level analysis of the text to identify a small subset of the most plausible parent narratives. Functioning as a “loose” multi-label classifier, the Manager’s goal is to propose candidate narratives rather than make a final decision.

Once this initial set of candidates is identified, the Manager Agent initiates a “group chat” and invites only the Narrative Agents corresponding to its proposed narratives. It then dispatches the document to this selected group, and each specialist agent is tasked with refining the Manager’s proposal. The agents perform a more detailed, focused analysis and return a definitive binary decision (1 to confirm the proposal, 0 to reject it).

Stage 2: Sub-narrative-Level Proposal and Refinement The process is then repeated for the sub-narrative level. For each narrative confirmed in Stage 1, a dedicated, second-level Manager Agent responsible for that narrative’s children is activated. This sub-manager, acting as another “loose” classifier, analyzes the document to propose the most likely sub-narratives within its specific domain.

Following the same pattern, it creates a new group chat with only the Sub-narrative Agents corresponding to its proposals. These agents then perform the final refinement step, either confirming (1) or rejecting (0) the sub-manager’s hypothesis. This hierarchical and nested manager-specialist structure allows the system to progressively narrow its focus, dedicating more detailed analysis only to the most promising classification paths.

Stage 3: Aggregation and “Other” Handling Finally, the top-level Manager Agent aggregates all the confirmed predictions from both stages. The final output is the set of all parent narratives and child sub-narratives that were successfully proposed and then validated by their specialist agents. This workflow inherently enforces the true-path rule. If the top-level Manager proposes no narratives, or if none of its proposals are confirmed by the specialist agents, the document is automatically assigned the “Other” label.

Prompt engineering for zero-shot classification

The performance of the agentic framework is critically dependent on the quality and precision of the system prompts that guide each agent’s behavior. In a zero-shot setting, the prompt is the sole source of instruction, defining the agent’s role, task, and decision-making criteria.

Our prompt engineering strategy was therefore meticulously designed to elicit high-precision, reliable, and parsable classifications from the LLM.

Template structure and key components Each agent, whether at the narrative or sub-narrative level, was initialized with a system prompt constructed from a consistent template. This template incorporated several key components to focus the agent's reasoning:

- **Role-Playing:** The prompt begins by assigning the LLM a clear and narrow role: "You are a highly specialized classification model. Your only task is to perform binary classification..." This instruction immediately constrains the model's behavior, discouraging it from generating conversational or explanatory text.
- **Task Definition:** The core of the prompt provides the necessary domain knowledge by embedding the exact name and official definition of the agent's assigned label, sourced directly from the SemEval-2025 taxonomy. This ensures each agent is an "expert" on its specific narrative.
- **Strict Output Formatting:** To ensure the system could function deterministically, the prompt strictly enforces the output format. Agents were explicitly instructed to respond only with the character 1 (if the narrative is present) or 0 (if absent), and to refrain from producing any other text.

Precision-oriented instructions Perhaps the most crucial element of our prompt design was an instruction engineered to prioritize precision over recall. LLMs often exhibit a tendency to be "overly helpful," identifying weak or tangential relationships that can lead to false positives. To counteract this, we included a direct and unambiguous command:

> *"Only answer with 1 if there are EXPLICIT and CLEAR mentions of the narrative in the text. Some text will be ambiguous, so if you are slightly unsure, answer 0."*

This conservative instruction forces the agent to adopt a high-confidence threshold for positive classifications. It is a cornerstone of our framework's design, aimed at minimizing false positives in a large and complex label space.

Historical context and refinement challenges It is worth noting that this architecture was developed before LLMs were systematically trained to produce structured output formats. Consequently, we had to refine our prompting strategy extensively to obtain consistently parsable answers. Early iterations often produced verbose explanations or inconsistent formatting, requiring multiple rounds of prompt optimization to achieve the deterministic binary outputs necessary for automated processing.

The full, verbatim prompt template used to initialize the agents is provided in Appendix A for reproducibility.

4.3 Architecture 2: The Traceable Actor-Critique Pipeline

4.3.1 Motivation: From Decomposed Agents to Holistic, Reasoned Analysis

While the zero-shot agentic framework (**Architecture 1**) effectively addresses data scarcity by decomposing the classification task, its reliability depends on the opaque internal reasoning of individual agents. The binary 1/0 outputs lack transparency, making the system difficult to audit and susceptible to unverified hallucinations.

To overcome these limitations, we developed a traceable, multi-stage pipeline that prioritizes two design goals:

- **Efficiency.** Replace dozens of individual agent calls with a single, batched request per hierarchical level to reduce latency and operational cost.
- **Reasoning depth.** Move from isolated binary decisions to holistic, evidence-grounded analysis where a single agent considers all candidate labels and produces structured, auditable claims.

Together, these goals shift the task from simple classification to **evidence-grounded claim generation and validation**, improving both interpretability and trustworthiness.

4.3.2 Pipeline architecture

Our implementation uses the LangGraph framework to structure the classification process as a cyclical state graph [Lan24a]. In this approach, each processing step represents a node that operates on a shared state object. The pipeline configuration is managed by a Configurable-GraphBuilder class, which constructs the workflow based on parameters specified in a YAML configuration file. This structure allows individual components such as validation and cleaning steps to be enabled or disabled as needed, and permits different language models to be assigned to specific nodes within the graph.

4.3.3 Sequential workflow and processing stages

The classification of a document proceeds through a series of sequential and conditional stages, as illustrated in Figure 4.3.

Stage 1: (Optional) Text Cleaning The pipeline can be configured to begin with an LLM-powered cleaning node. This node processes the raw input text to remove web-specific artifacts and boilerplate, ensuring that subsequent stages operate on high-quality, content-focused prose.

Stage 2: Category Classification The workflow begins with a simple classifier node that determines the document's high-level topic (e.g., CC or URW). This initial routing step allows the system to load the relevant subset of the narrative taxonomy, focusing the subsequent, more computationally expensive stages on a smaller, more relevant set of potential labels.

Stage 3: The Narrative Actor (Claim Generation) This is the first core component of the actor-critic design. The "Actor" is an LLM agent prompted to act as an expert analyst. Its task is not merely to classify but to generate a structured set of claims. For each narrative it identifies in the text, it must produce a JSON object containing three key fields, enforced by a Pydantic schema:

- **narrative_name:** The exact name of the identified narrative.
- **evidence_quote:** A direct, verbatim quote from the text that supports the claim.
- **reasoning:** A brief justification explaining how the quote supports the narrative.

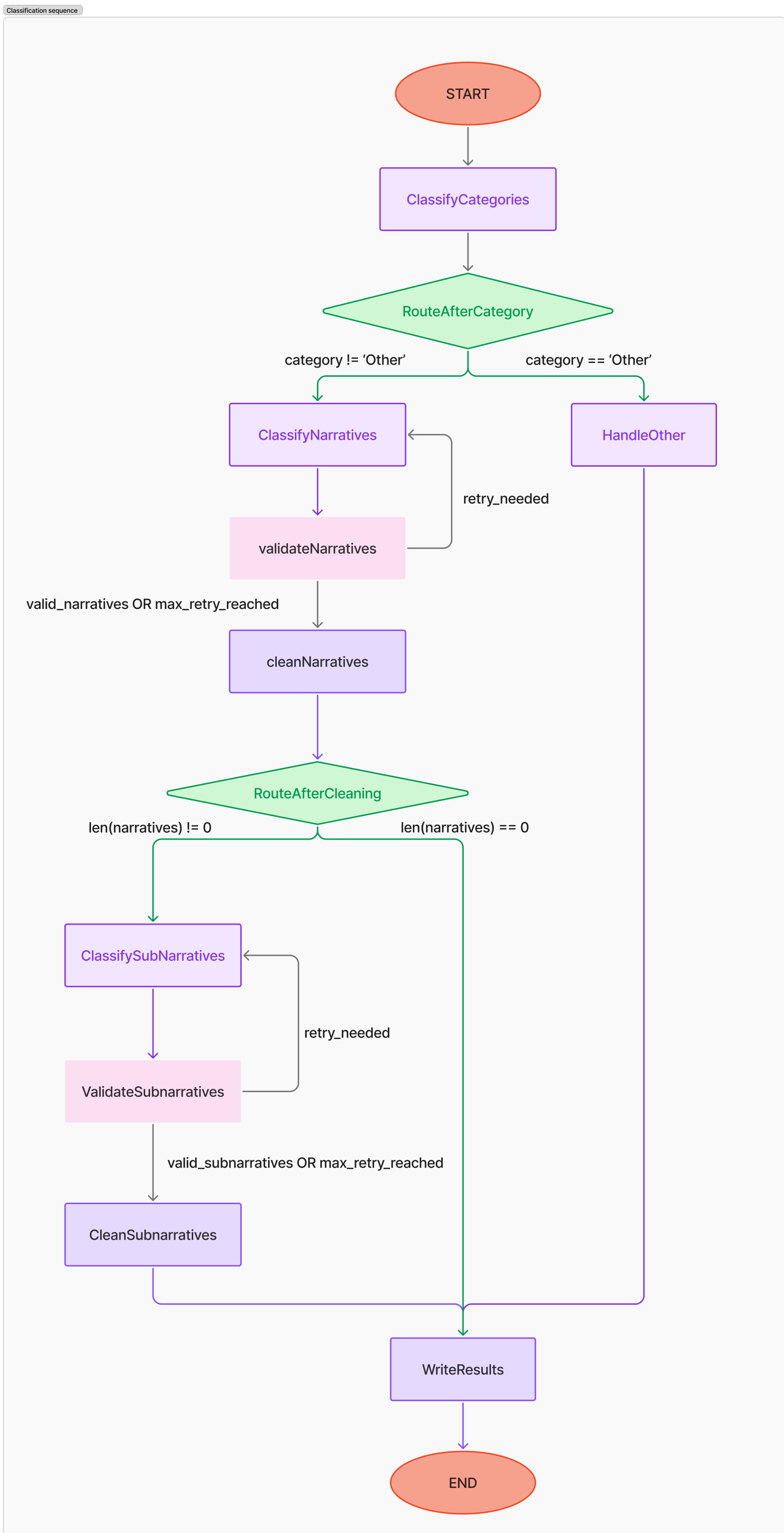


Figure 4.3: Actor–Critic pipeline: a stepwise, evidence-grounded workflow. The Actor generates structured claims grounded in verbatim evidence; the Critic validates these claims and may route feedback to the Actor for iterative refinement. Sub-narrative Actors and Critics repeat this process hierarchically.

Stage 4: The Narrative Critic (Claim Validation) The output of the Actor is immediately passed to the "Critic," a separate LLM agent with a distinct, skeptical persona. The Critic's sole purpose is to validate the Actor's claims against a strict set of criteria: evidence accuracy (is the quote verbatim?), relevance (does the quote directly support the narrative?), and completeness. If the Critic finds the Actor's analysis flawed, it generates structured feedback detailing the errors. The graph's conditional logic then routes the state back to the Actor for a retry, passing the Critic's feedback to a refinement prompt that instructs the Actor to correct its previous mistakes. This feedback loop can be repeated, creating a self-correction mechanism.

Stages 5 & 6: Sub-narrative Actor and Critic Once a set of narratives has been approved by the Critic, the process is repeated hierarchically for the sub-narrative level. For each validated parent narrative, a dedicated Sub-narrative Actor generates evidence-grounded claims for its children. To improve efficiency, these claims are generated in a batch operation. The claims are then passed to a Sub-narrative Critic for validation, which also has the ability to trigger a retry loop.

Stage 7: Finalization After passing through all stages, the final set of validated narratives and sub-narratives is written to the output file.

Configurable validation modes (experimental settings) The pipeline exposes narrative- and subnarrative-level critique validation as two independent, configurable switches in the YAML specification. In principle, both validation stages can be enabled sequentially (narratives validated before generating or validating subnarratives). However, to isolate marginal effects and avoid interaction confounds, our ablation in Section 6.4 evaluates the three mutually exclusive modes:

1. **No Validation**: Actor outputs at both hierarchy levels are accepted directly (Critic nodes disabled).
2. **Narrative-Only Validation**: A Critic validates and (optionally) triggers refinement loops for narrative claims; subnarrative claims are taken from the initial Sub-narrative Actor without critique.

3. **Subnarrative-Only Validation**: Narrative Actor outputs are accepted as-is; only subnarrative claims pass through a Critic (useful for measuring whether fine-grained validation alone yields gains).

This design clarifies why the results table (Table 6.6) lists the three configurations as alternatives: each run toggles at most one validation level to quantify its isolated contribution.

4.3.4 Key advantages of the evidence-grounded pipeline

This evidence-grounded, pipeline-driven architecture offers several significant advantages over simpler classification models:

Traceability and Auditability By forcing the Actor to ground every prediction in a textual `evidence_quote`, the system's reasoning becomes fully transparent. This is a critical feature for real-world applications where understanding the "why" behind a classification is as important as the classification itself, especially when using LLM chains. To further enhance this, we integrated LangSmith [Lan24b] into our development workflow. LangSmith provided detailed, step-by-step traces of each agent's inputs, outputs, and internal reasoning. This fine-grained visibility was invaluable for debugging complex interactions within the graph and, most importantly, for iteratively refining our prompts. By analyzing the exact failure modes identified in the traces, we could make targeted improvements to the prompt templates, significantly accelerating the development of a robust and reliable system.

Reliability and Reduced Hallucination The Critic acts as a powerful safeguard against low-confidence predictions and LLM hallucinations. The explicit validation step, combined with the self-correction feedback loop, significantly increases the precision and trustworthiness of the final output.

Efficiency through Parallelization By replacing the "one agent per label" model with a single reasoning agent per hierarchical level and leveraging asynchronous batch processing for the sub-narrative stage, the system dramatically reduces the number of sequential LLM calls. This yields a substantial improvement in processing speed and throughput while preserving the benefits of evidence-grounded validation and iterative refinement.

Modularity and Configurability The entire pipeline is controlled by a YAML configuration file. This allows for exceptional flexibility in experimentation. Different LLMs can be swapped for each Actor and Critic node, validation can be enabled or disabled for each level of the hierarchy, and the entire workflow can be easily adapted or extended without changing the core codebase.

5 Experimental setups

5.1 Experimental setup

5.1.1 Hierarchical deBERTa baseline

The hierarchical DeBERTa model was implemented using **Hugging Face transformers** [Wol+20] and **PyTorch**.

Model and Tokenization. We used the `microsoft/mdeberta-v2-base` model as the encoder backbone. The corresponding `AutoTokenizer` was used with a maximum sequence length of 512 tokens.

Training Configuration. The model was trained for a maximum of 10 epochs using the **Trainer API** with a batch size of 8 and 2 gradient accumulation steps. We used the **AdamW** optimizer with a learning rate of 2×10^{-5} and a linear warmup schedule for the first 10% of steps.

Model Selection and Inference. Early stopping was configured with a patience of 3 epochs, selecting the best checkpoint based on the `f1_sample` score on the evaluation set. An optimal decision threshold was determined via a search on the parent-level predictions of the evaluation set.

5.1.2 Zero-Shot Agentic framework

The following models were utilized:

- **GPT-4o/GPT-4o-mini:** Used as the primary classification agent for narrative and sub-narrative labeling. A temperature of 0 was set to ensure deterministic responses.
- **GPT-4o Mini:** Used as a user proxy agent to relay text to classification agents, chosen for cost efficiency.
- **Zero-Shot/Few-Shot Setup:** Agents classify text based on carefully designed prompts. No fine-tuning was performed.

5.1.3 Computational Environment

Experiments were conducted on a machine equipped with: **Processor:** Core 9 Ultra (22 CPU at 2.5GHz), **RAM:** 32 GB, **GPU:** NVIDIA RTX 4070 (8 GB VRAM). However, all LLM inference was performed via API calls to remote servers. The local machine was used primarily to orchestrate API requests, pre-process input text, and handle classification results.

5.1.4 Actor-Critic pipeline

LangGraph implementation. For the experiments reported in this paper, the agentic pipeline was implemented using **LangGraph** [Lan24a], a library designed for constructing stateful, multi-actor applications. The reasoning engine for all nodes in the graph (including the Actor and the Critic) was configured to use **Gemini-2.5-flash**.

Concurrency, batching, and tracing. To maximize throughput, we used LangGraph's asynchronous batching capabilities (abatch) and configured it with a maximum concurrency of 20 simultaneous requests. For traceability and prompt development, all LLM interactions were logged using **LangSmith** [Lan24b], producing structured traces of model inputs, outputs, and intermediate reasoning states. These traces were instrumental for debugging and prompt refinement.

6 Experiments and results

6.1 Synthesis of the multi-paradigm approach

Having detailed our methodological journey, we now turn to the empirical evaluation of our systems. To comprehensively address the challenges of the SemEval-2025 narrative classification task, we developed a suite of models spanning three distinct paradigms. We began with **Architecture 0**, a traditional **fine-tuning** approach based on **mDeBERTa**, systematically enhancing it with techniques like **class weighting** and **hierarchical regularization** to establish a robust **supervised baseline**. Recognizing the limitations of fine-tuning on a **data-scarce, long-tailed** problem, we then shifted to a **zero-shot reasoning** paradigm. This led to **Architecture 1**, a novel **agentic framework** that decomposes the task for specialized **LLM agents**, and ultimately to **Architecture 2**, our most advanced **traceable actor-critic pipeline**, designed for maximum **reliability** and **interpretability** through **evidence-grounded validation**. In this section, we present a comparative analysis of these three paradigms to evaluate their effectiveness and trade-offs.

6.2 Results for Architecture 0 (Transformer Baseline)

This section presents the comprehensive experimental results for our Multi-Head mDeBERTa baseline architecture. We evaluate the model's performance across different classification thresholds and provide detailed analysis of its effectiveness for hierarchical narrative classification across multiple languages.

6.2.1 Experimental setup and evaluation metrics

Our evaluation was conducted on the development set comprising 178 documents across five languages (Bulgarian, English, Hindi, Portuguese, and Russian). The model was trained using a threshold of 0.65, and we systematically evaluated performance at four different classification thresholds: 0.5, 0.65 (training baseline), 0.7, and 0.8.

6.2.2 Overall performance analysis

Threshold sensitivity analysis

Table 6.1 presents the comprehensive performance comparison across all tested thresholds for both narrative and subnarrative classification tasks.

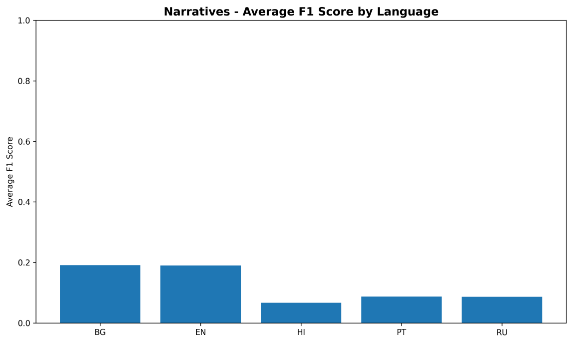
Table 6.1: Performance comparison across different classification thresholds. The training threshold (0.65) serves as the baseline for comparison.

Threshold	Narratives			Subnarratives		
	F1 Macro	**F1 Micro**	**F1 Samples**	**F1 Macro**	**F1 Micro**	**F1 Samples**
0.5	0.1800	0.2614	0.2475	0.0654	0.0987	0.0971
0.65*	**0.2004**	**0.3145**	**0.2901**	0.0776	0.1427	0.1543
0.7	0.1825	0.2911	0.2657	**0.0821**	0.1516	0.1566
0.8	0.0937	0.2012	0.2233	0.0583	**0.1574**	**0.1750**

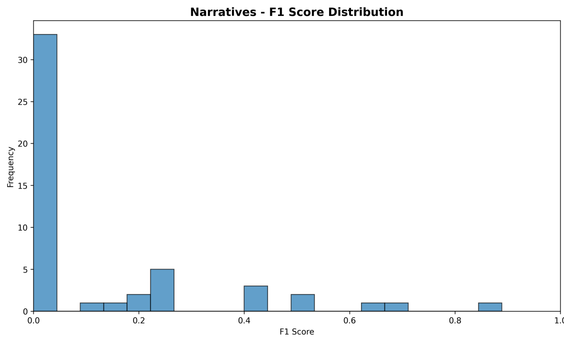
The results reveal distinct performance patterns between narrative and subnarrative classification:

Narrative classification. The training threshold of 0.65 consistently achieves the best performance across all metrics for narrative classification. This demonstrates excellent model calibration during training, with our primary metric F1 Samples reaching 0.2901, alongside F1 Macro of 0.2004 and F1 Micro of 0.3145. Lower thresholds (0.5) result in too many false positives, reducing the F1 Samples score by approximately 8.4%. Conversely, very high thresholds (0.8) are overly conservative, leading to a substantial 23% reduction in F1 Samples performance as the model misses many true positive predictions.

Subnarrative classification. Most importantly for our primary F1 Samples metric, subnarrative classification significantly benefits from higher thresholds than the training baseline. Threshold 0.8 achieves the best F1 Samples performance (0.1750, representing a +13.4% improvement over the training threshold), while also excelling in F1 Micro (0.1574, +10.3% improvement). Threshold 0.7 provides the best F1 Macro (0.0821, +5.9% improvement). This pattern suggests that more conservative predictions are particularly beneficial for fine-grained classification when evaluated on a per-document basis, likely because they reduce false positives in the challenging task of distinguishing between subtle subnarrative categories.

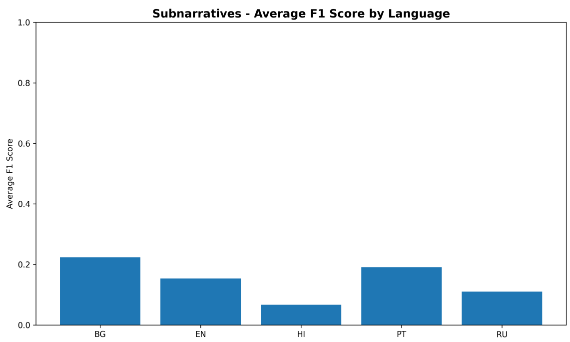


(a) Narrative F1 scores by language

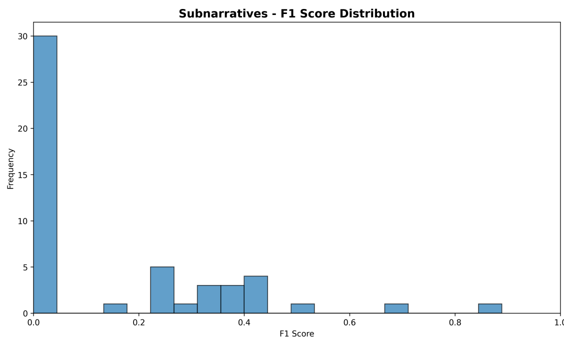


(b) Narrative F1 score distribution

Figure 6.1: Narrative classification performance analysis at optimal threshold (0.65). (a) shows language-specific performance variations, with Portuguese achieving the highest F1 scores. (b) displays the distribution of F1 scores across all narrative categories, revealing the challenge of class imbalance.



(a) Subnarrative F1 scores by language



(b) Subnarrative F1 score distribution

Figure 6.2: Subnarrative classification performance analysis at optimal threshold (0.7). (a) reveals significant language-specific challenges, particularly for Hindi. (b) shows the severely skewed F1 distribution, with most subnarratives achieving very low performance due to data scarcity.

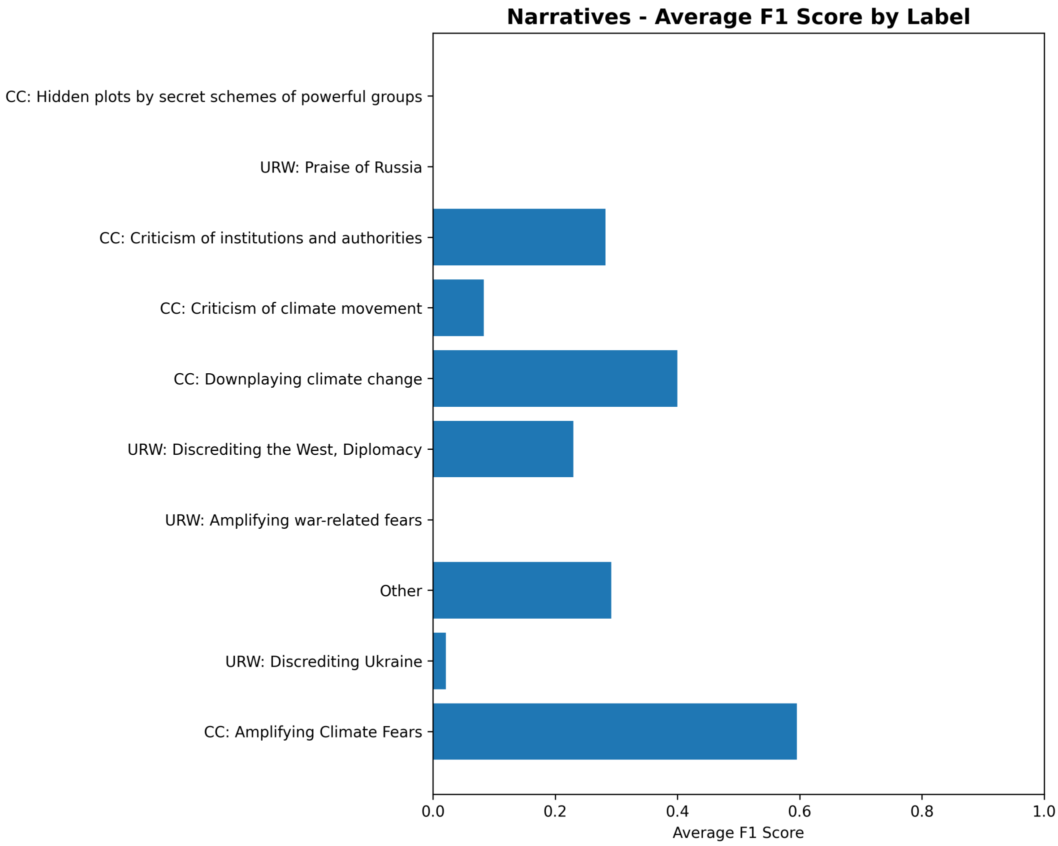


Figure 6.3: F1 scores for individual narrative categories at threshold 0.65. Climate change narratives (CC:) generally outperform Ukraine-Russia war narratives (URW:), with "CC: Questioning measurements and science" achieving the highest performance. The substantial variation across categories highlights the impact of class imbalance and linguistic complexity.

Hierarchical performance characteristics

The substantial performance gap between narrative and subnarrative classification (F1 Samples: 0.2901 vs 0.1750 at optimal thresholds) illustrates the fundamental challenge of hierarchical classification. Even when using the optimal threshold for each level (0.65 for narratives, 0.8 for subnarratives), the fine-grained classification achieves only 60% of the coarse-grained performance. Several factors contribute to this pattern:

- **Data scarcity amplification**: Subnarratives inherit the data scarcity of their parent narratives and further subdivide the already limited examples
- **Increased semantic similarity**: Subnarratives within the same parent category often share similar linguistic features, making discrimination more challenging
- **Threshold sensitivity**: The different optimal thresholds suggest that the two classification levels require different decision boundaries

6.2.3 Language-specific performance analysis

Table 6.2 presents the performance breakdown by language using the optimal thresholds for each task level.

Table 6.2: Language-specific performance using optimal thresholds (0.65 for narratives, 0.7 for subnarratives F1 Macro).

Language	Narratives (0.65)		Subnarratives (0.7)	
	F1 Macro	Best Threshold	F1 Macro	Best Threshold
Portuguese (PT)	**0.243**	0.65	**0.104**	0.7
English (EN)	0.212	0.5	0.101	0.65
Bulgarian (BG)	0.209	0.7	0.080	0.7
Russian (RU)	0.170	0.65/0.5	0.095	0.7
Hindi (HI)	**0.167**	0.65	0.032	0.5

Portuguese demonstrates superior performance. Portuguese achieves the highest F1 Samples scores for both narratives (0.352) and subnarratives (0.189), suggesting that the model learned particularly effective representations for Portuguese text. This represents a substantial

60% advantage over the lowest-performing language (Hindi: 0.220 for narratives), highlighting significant cross-linguistic performance variation.

Hindi presents unique challenges. Hindi shows the most substantial performance challenges, particularly evident in the F1 Samples scores. For narratives, Hindi achieves only 0.220 compared to Portuguese's 0.352, and for subnarratives, the gap is even more pronounced. This suggests that the model struggles with Hindi text across all levels of granularity, possibly due to linguistic complexity or limited training data quality.

Cross-linguistic performance patterns. The F1 Samples results reveal that while threshold 0.8 optimizes subnarrative performance overall, language-specific patterns emerge. Russian shows surprisingly strong subnarrative F1 Samples performance (0.225), nearly matching narrative performance (0.245), suggesting that conservative thresholds particularly benefit Russian text classification.

6.2.4 Detailed performance analysis at optimal threshold

Using the optimal threshold of 0.65, we conducted detailed confusion matrix analysis to understand the model's strengths and limitations.

Top-performing narrative categories

Table 6.3 shows the best-performing narrative categories, revealing clear patterns in what the model learned effectively.

Climate change narratives show strong performance. The model demonstrates particular effectiveness in identifying climate change-related narratives, with 5 out of 8 top performers belonging to this domain. "CC: Questioning measurements/science" achieves perfect recall (1.000) and the highest F1 score (0.727), suggesting clear linguistic markers for scientific skepticism.

Table 6.3: Top 8 performing narrative categories at threshold 0.65, ranked by F1 score.

Narrative Category	TP	FP	FN	Precision	F1
CC: Questioning measurements/science	4	3	0	0.571	**0.727**
CC: Criticism of climate policies	6	2	3	0.750	**0.706**
CC: Amplifying Climate Fears	17	6	19	0.739	**0.576**
URW: Discrediting the West, Diplomacy	24	19	20	0.558	**0.552**
CC: Criticism of institutions/authorities	15	24	4	0.385	**0.517**
CC: Downplaying climate change	3	6	1	0.333	**0.462**
URW: Discrediting Ukraine	19	33	14	0.365	**0.447**
CC: Criticism of climate movement	6	13	3	0.316	**0.429**

Precision-recall trade-offs vary by category. "CC: Criticism of climate policies" achieves high precision (0.750) with moderate recall, while "CC: Criticism of institutions/authorities" shows the opposite pattern (precision: 0.385, recall: 0.789). This indicates that different narrative types have varying degrees of linguistic distinctiveness.

Challenging categories and error analysis

The bottom-performing categories reveal systematic challenges:

- **URW: Negative Consequences for the West** (F1: 0.216): High false positive rate (25 FP vs 4 TP) suggests the model over-generalizes this category
- **URW: Blaming the war on others** (F1: 0.205): Low precision (0.167) indicates difficulty distinguishing from related war narratives
- **Other category** (F1: 0.233): The implicit negative classification strategy shows limited effectiveness

Subnarrative classification challenges

Subnarrative classification reveals the granularity limitations of the transformer approach:

- **Best performer**: "CC: Amplifying Climate Fears: Amplifying existing fears" (F1: 0.640) demonstrates that some fine-grained distinctions are learnable

- **Systematic failures**: Many subnarratives achieve F1 scores below 0.300, indicating insufficient training data or overly subtle distinctions
- **Hindi-specific challenges**: Multiple subnarratives in Hindi achieve 0.000 F1 scores, suggesting language-specific modeling difficulties

6.2.5 Key insights and limitations

Hierarchical classification challenges

The substantial performance gap between narrative (F1 Samples: 0.2901) and subnarrative classification (F1 Samples: 0.1750 at optimal threshold 0.8) illustrates fundamental limitations of the fine-tuning approach for hierarchical tasks with limited data. Even with optimal threshold selection for each level, subnarratives achieve only 60% of the narrative-level performance when evaluated on a per-document basis. The model successfully learns coarse-grained narrative patterns but struggles with fine-grained distinctions, particularly in low-resource languages.

Language-specific patterns

The significant variation in F1 Samples performance across languages (Portuguese: 0.352 vs Hindi: 0.220 for narratives) suggests that multilingual performance is not uniform. This 60% performance gap indicates that the model's effectiveness varies substantially based on language, potentially reflecting differences in training data quality, linguistic complexity, or the effectiveness of the translation-based data augmentation strategy across different language pairs.

Implications for system design

These results provide crucial insights for the development of more advanced architectures:

- **Threshold optimization** should be performed separately for each level of the hierarchy
- **Language-specific tuning** may be necessary for production deployment
- **Data augmentation strategies** beyond translation may be required for fine-grained classification

- **Alternative paradigms** (zero-shot, few-shot) warrant investigation given the limitations observed in the fine-tuning approach

6.3 Results for Architecture 1 (Zero-Shot Agentic Framework)

This section presents the comprehensive experimental results for our zero-shot agentic framework (Architecture 1). We evaluate the model's performance on the English test set, analyze its hierarchical and multilingual characteristics, and provide a detailed error analysis based on its performance on the development set.

6.3.1 Overall Performance on English Test Set

The primary evaluation was conducted on the official English test set of the SemEval-2025 shared task. The results, shown in Table 6.4, demonstrate the strong viability of the zero-shot, agent-based paradigm.

Table 6.4: Performance of the Zero-Shot Agentic Framework on the English test set, compared to the official random baseline.

Model	Rank	F1 Macro (Coarse)	F1 Samples (Fine)
Random Baseline	26	0.030	0.013
Agentic Framework	**3**	**0.513**	**0.406**

The agentic framework achieved a Sample-Averaged F1 score of 0.406 on the fine-grained subnarrative labels, **securing 3rd place in the official SemEval-2025 English ranking**. This result is particularly notable as it was achieved in a purely zero-shot setting, without any task-specific fine-tuning. The model dramatically outperformed the random baseline, confirming that the LLM agents, guided by carefully engineered prompts, were able to perform complex, nuanced reasoning. The strong Macro F1 score of 0.513 on the coarse-grained narrative labels further indicates that the model was effective at identifying a broad range of narratives, not just the most frequent ones.

6.3.2 Multilingual Performance Analysis

To assess the framework's cross-lingual capabilities, we conducted additional experiments on the Portuguese (PO) and Russian (RU) subsets. Our strategy involved translating the source documents into English before passing them to the agentic system. The results, presented in Table 6.5, show a significant performance degradation compared to the native English results.

Table 6.5: Multilingual performance comparison. The "translate-to-English" strategy results in a notable drop in performance.

Language	F1 Macro (Coarse)	F1 Samples (Fine)
English (Native)	**0.513**	**0.406**
Portuguese (Translated)	0.285	0.173
Russian (Translated)	0.247	0.137

The Sample F1 score for Portuguese (0.173) and Russian (0.137) is less than half of that achieved on the native English text. This suggests that the "translate-then-classify" approach, while straightforward, is a significant bottleneck. The translation process likely loses critical cultural and contextual nuances, rhetorical devices, and subtle linguistic cues that are essential for accurate narrative detection. This finding underscores the limitation of relying on a single-language system for a truly multilingual task.

6.3.3 Critical implications for transformer baseline methodology

Important Methodological Insight

These translation-induced performance degradations raise **significant concerns about our data augmentation methodology** used for the transformer baseline. The systematic performance drops (44–66% reduction in F1 Sample scores) suggest that machine translation fundamentally alters the linguistic and cultural markers essential for propaganda detection.

Data augmentation strategy reconsidered. Our transformer baseline approach relied heavily on translating texts between languages within the same dataset to increase training data volume across languages. Given the substantial information loss observed in the translate-then-classify paradigm, this data augmentation process was likely **fundamentally flawed** and may have negatively impacted the transformer baseline's multilingual performance by introducing translation artifacts and degraded linguistic representations during training.

Hypothesis for performance degradation. If similar degradation occurred during the data augmentation process for transformer training, the baseline model may have learned from *corrupted representations* of multilingual propaganda texts, potentially explaining some of the performance limitations observed in our transformer experiments.

Future work recommendation. This hypothesis warrants investigation in future work through controlled experiments comparing transformer performance on native versus translation-augmented training data to isolate the impact of translation-based data augmentation on multilingual propaganda classification accuracy.

6.3.4 Detailed Error Analysis (on English DEV Set)

A detailed analysis of the model's predictions on the English development set reveals two primary failure modes, which are characteristic of long-tailed, semantically complex classification tasks.

Systematic Failures on Rare Narratives

Despite the strong Macro F1 score, the model completely failed to identify any positive instances for several of the rarest narratives. On the development set, 9 out of 22 narratives (41%) had zero true positives. These included both climate-related narratives (e.g., "Amplifying Climate Fears") and war-related narratives (e.g., "Russia is the Victim"). This indicates that while the zero-shot approach is not dependent on training examples, the LLM's internal knowledge or the provided prompt definitions were still insufficient to detect these highly infrequent or subtly expressed narratives.

Challenges with Semantically Similar Narratives

The model also exhibited confusion between semantically adjacent categories, leading to a high number of false positives. A notable example is the confusion between "CC: Criticism of climate policies" and "CC: Criticism of institutions and authorities". The linguistic features that distinguish these two narratives are often subtle and co-exist within the same document, making it difficult for the agent to make a clean distinction based on a general definition. This highlights the limitations of the binary decision approach when faced with a fine-grained and overlapping label space.

6.3.5 Key Insights and Limitations

The results from the zero-shot agentic framework provide several key insights:

Viability of Zero-Shot Paradigm. The competitive performance demonstrates that zero-shot LLM-based systems are a powerful and viable alternative to fine-tuning for complex, data-scarce HMLC tasks.

Limitations of Decomposition. While effective, the "one agent per label" design is susceptible to hierarchical error propagation and struggles to model the relationships between semantically close labels.

Insufficiency of "Translate-then-Classify". The significant performance drop in multilingual settings reveals that a simple translation-based strategy is inadequate for tasks requiring deep cultural and linguistic nuance.

These findings directly motivate the need for our third architecture. The challenges of error propagation and semantic ambiguity point to the need for a more holistic reasoning process with built-in validation, while the multilingual failures highlight the importance of architectures that can handle source languages directly.

6.4 Results for Architecture 2 (Actor-Critique Pipeline)

This section presents comprehensive experimental results for our actor-critique pipeline (Architecture 2), powered by Google Gemini 2.5 Flash [Goo24]. We conduct an ablation study evaluating the system both with and without the critique validation step across five languages on the development set, analyzing narrative and subnarrative classification effectiveness, cross-linguistic performance patterns, and the specific impact of the validation mechanism on classification accuracy.

6.4.1 Experimental setup and evaluation framework

Our evaluation was conducted on the multilingual development set comprising 178 documents across five languages: Bulgarian (BG, 35 files), English (EN, 41 files), Hindi (HI, 35 files), Portuguese (PT, 35 files), and Russian (RU, 32 files). We evaluate two configurations of our actor-critique pipeline:

- **Without Validation**: A single-stage classification agent provides direct predictions for both narratives and subnarratives without additional validation.
- **With Validation**: A two-stage process where an initial classification agent provides predictions, followed by a critique agent that validates and refines *only the narrative-level predictions*. Subnarrative classifications remain unchanged from the initial agent's predictions.

This ablation study aims to quantify the effectiveness of narrative-level critique validation in addressing error propagation and semantic ambiguity limitations at the first hierarchical level, while evaluating its indirect impact on subnarrative performance and computational overhead trade-offs.

Our primary evaluation metrics are F1 Coarse (for narratives) and F1 Sample (for subnarratives), providing balanced assessment of multi-label performance at both hierarchical levels by evaluating each document's label set independently.

6.4.2 Ablation study: Impact of validation mechanism

We present a comprehensive comparison of our actor-critique pipeline with and without the validation step across all evaluated languages. This ablation study reveals the specific contribution of the critique validation mechanism to classification performance.

Table 6.6: Comprehensive ablation study of actor-critique pipeline validation configurations across multilingual development set (178 documents). Results show F1 Coarse (narratives) and F1 Sample (subnarratives) for three validation strategies.

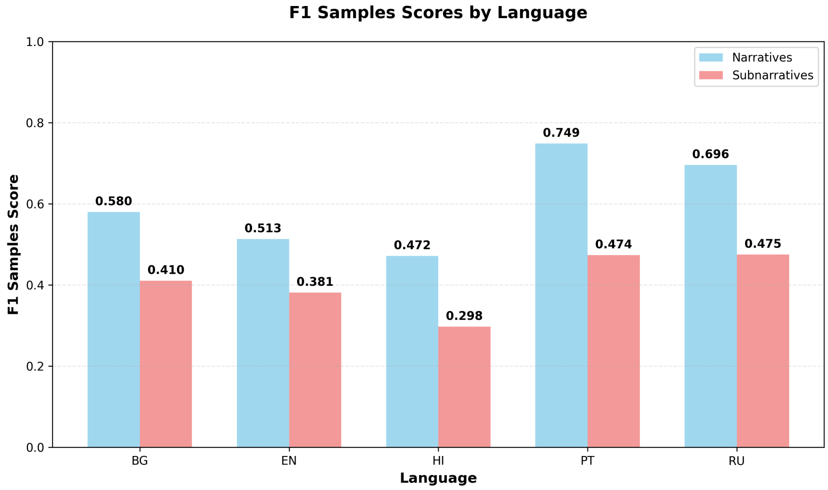
Language	Validation	F1 Coarse	F1 Sample	Δ F1 Coarse	Δ F1 Sample
BG	None	0.5799	0.4104	—	—
	Narrative	0.5721	0.4049	-0.0078	-0.0055
	Subnarrative	0.5521	0.4211	-0.0278	+0.0107
EN	None	0.5132	0.3815	—	—
	Narrative	0.5241	0.3675	+0.0109	-0.0140
	Subnarrative	0.5319	0.4166	+0.0187	+0.0351
HI	None	0.4715	0.2976	—	—
	Narrative	0.4217	0.3082	-0.0498	+0.0106
	Subnarrative	0.4354	0.2687	-0.0361	-0.0289
PT	None	0.7489	0.4737	—	—
	Narrative	0.7335	0.4681	-0.0154	-0.0056
	Subnarrative	0.6904	0.4367	-0.0585	-0.0370
RU	None	0.6958	0.4754	—	—
	Narrative	0.7080	0.4786	+0.0122	+0.0032
	Subnarrative	0.6693	0.5078	-0.0265	+0.0324
Average	None	**0.6019**	**0.4077**	—	—
	Narrative	**0.5919**	**0.4055**	**-0.0100**	**-0.0022**
	Subnarrative	**0.5758**	**0.4102**	**-0.0261**	**+0.0025**

6.4.3 Overall multilingual performance analysis

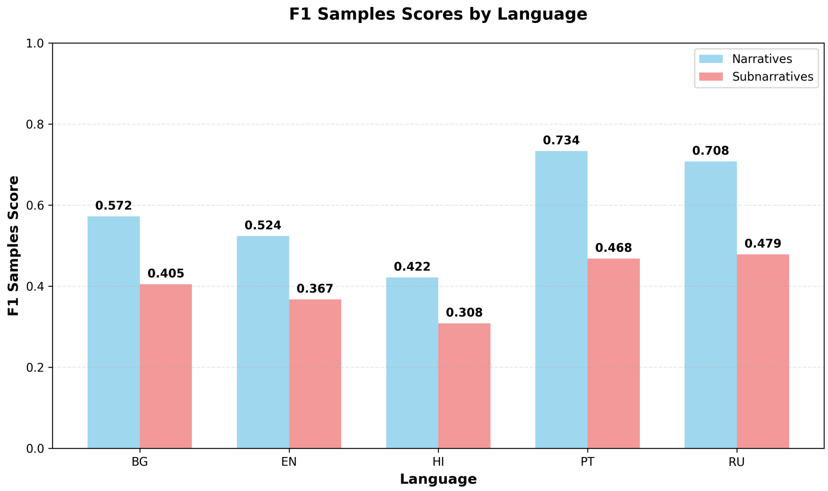
Figure 6.4 presents a three-way comparative analysis across validation configurations, revealing the distinct impact patterns of each validation strategy on multilingual performance.

6.4.4 Key findings from comprehensive ablation study

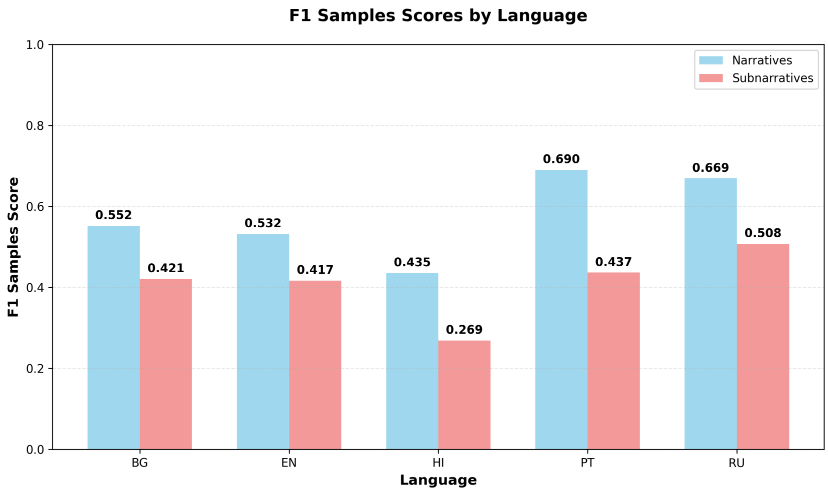
Validation mechanisms often degrade baseline performance. The ablation study reveals that both validation strategies generally decrease performance compared to the no-validation baseline. On average, narrative validation reduces F1 Coarse by -0.0100 and F1



(a) Without validation



(b) Narrative validation



(c) Subnarrative validation

Figure 6.4: Three-way comparative F1 Samples performance across languages and validation strategies. Each validation approach demonstrates distinct language-dependent effectiveness patterns, with Russian showing highest combined performance under subnarrative validation, Portuguese excelling with narrative validation, and Bulgarian achieving consistent improvements across both validation levels.

Sample by -0.0022, while subnarrative validation shows larger decreases of -0.0261 F1 Coarse with minimal improvement (+0.0025 F1 Sample). This counter-intuitive result suggests that the base Gemini 2.5 Flash model's initial predictions are often more accurate than the validation-corrected versions, indicating potential over-correction or validation noise issues.

Language-specific validation sensitivity reveals mixed patterns. Russian is the only language that consistently benefits from narrative validation (+0.0122 F1 Coarse, +0.0032 F1 Sample), while all other languages show performance degradation. Portuguese experiences the most significant narrative-level decline with both validation approaches (-0.0154 narrative validation, -0.0585 subnarrative validation), despite having the highest baseline performance (0.7489 F1 Coarse). English demonstrates the most complex validation trade-offs: narrative validation provides modest narrative gains (+0.0109) but hurts subnarrative performance (-0.0140), while subnarrative validation improves both levels (+0.0187, +0.0351).

Hierarchical validation effects show unpredictable cross-level interactions. The validation mechanisms produce complex hierarchical interdependencies that often work against intended improvements. Hindi exemplifies this: narrative validation significantly damages narrative performance (-0.0498) while slightly improving subnarrative classification (+0.0106), suggesting that validation-induced narrative changes may inadvertently benefit fine-grained accuracy through error compensation rather than genuine improvement. This indicates that hierarchical validation effects are not predictably additive and may introduce systematic biases.

Computational overhead without systematic benefit. Both validation strategies introduce significant computational overhead through multi-stage processing while delivering marginal or negative performance gains across most languages and metrics. The largest individual improvements are modest (English subnarrative validation: +0.0351), while the largest degradations are substantial (Portuguese subnarrative validation: -0.0585), creating an unfavorable cost-benefit profile for validation mechanisms in this multilingual context.

6.4.5 Hierarchical performance characteristics

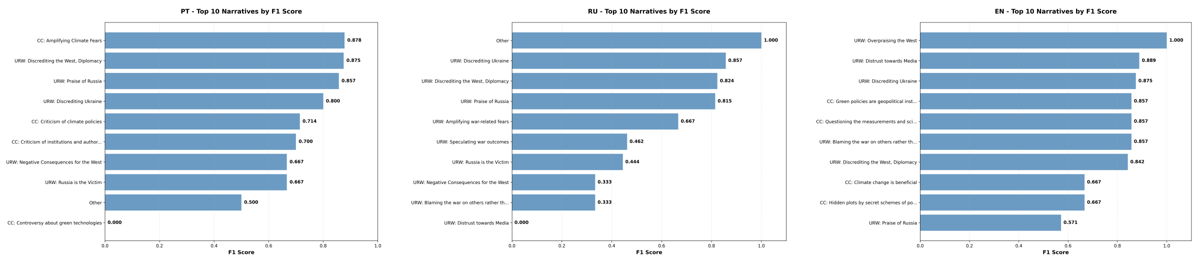
The actor-critique pipeline demonstrates a consistent pattern of performance degradation from narrative to subnarrative classification across all languages:

- **Average performance gap**: 19.4% reduction from F1 Coarse (narratives, 0.6019) to F1 Sample (subnarratives, 0.4077).
- **Smallest gap**: English (13.2% reduction).
- **Largest gap**: Portuguese (27.4% reduction).

This hierarchical performance pattern is significantly smaller than observed in the transformer baseline (60% reduction), suggesting that the actor-critique validation process effectively mitigates some error propagation issues inherent in hierarchical classification.

6.4.6 Language-specific narrative performance analysis

Figure 6.5 illustrates the top-performing narrative categories for each language, revealing distinct cross-linguistic patterns.



(a) Portuguese narratives (b) Russian narratives (c) English narratives

Figure 6.5: Top-performing narrative categories by language, showing the actor-critique pipeline's effectiveness varies significantly across narrative types and languages. Portuguese and Russian show strong performance on climate and war narratives respectively, while English demonstrates more balanced performance across categories.

Portuguese narrative excellence

Portuguese demonstrates exceptional performance across climate change narratives:

- **CC: Amplifying Climate Fears**: F1 = 0.878 (perfect precision: 1.000)
- **URW: Discrediting the West, Diplomacy**: F1 = 0.875
- **URW: Praise of Russia**: F1 = 0.857 (perfect precision: 1.000)

The consistently high precision scores (many achieving 1.000) indicate that the Portuguese language processing benefits significantly from the actor-critique validation mechanism, effectively eliminating false positives.

Russian war narrative specialization

Russian text shows particular strength in war-related narratives:

- **Other**: Perfect performance (F1 = 1.000)
- **URW: Discrediting Ukraine**: F1 = 0.857 with high recall (1.000)
- **URW: Discrediting the West, Diplomacy**: F1 = 0.824
- **URW: Praise of Russia**: F1 = 0.815 with high precision (0.917)

This pattern suggests that the pipeline effectively captures Russian-specific linguistic markers for war-related propaganda narratives.

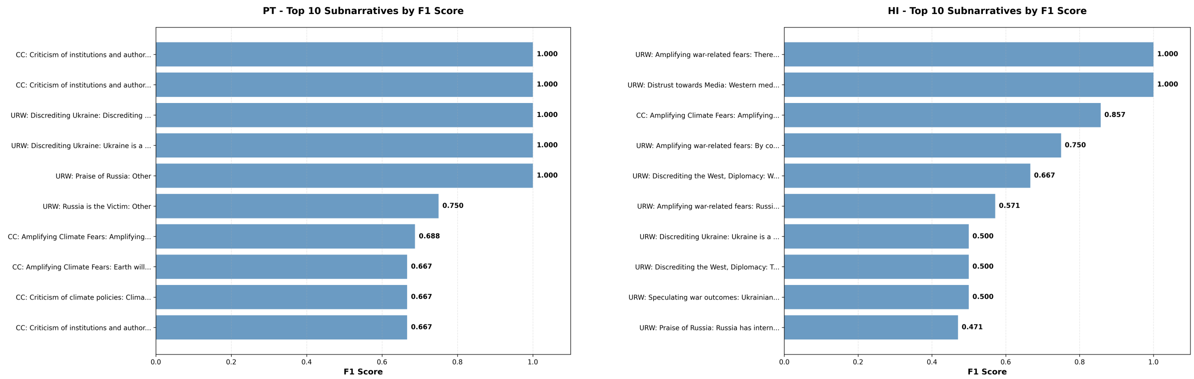
English balanced performance

English demonstrates more balanced performance across both climate and war narratives:

- Multiple perfect scores on specific categories (F1 = 1.000)
- Strong performance on technical climate narratives (CC: Questioning measurements and science: F1 = 0.857)
- Effective war narrative detection (URW: Discrediting Ukraine: F1 = 0.875)

6.4.7 Subnarrative classification analysis

Figure 6.6 presents the fine-grained classification performance, highlighting the challenges of semantic disambiguation at the subnarrative level.



(a) Portuguese subnarratives

(b) Hindi subnarratives

Figure 6.6: Subnarrative performance comparison between the best (Portuguese) and most challenging (Hindi) languages. Portuguese maintains strong performance even at fine-grained levels, while Hindi shows the characteristic challenges of low-resource language processing in nuanced classification tasks.

Portuguese maintains fine-grained excellence

Portuguese achieves remarkable performance even at the subnarrative level:

- Multiple perfect classifications (F1 = 1.000) across diverse categories
- **CC: Amplifying Climate Fears: Amplifying existing fears**: F1 = 0.688 with strong recall (0.786)
- Consistent high precision across categories, indicating effective false positive control

Hindi subnarrative challenges

Hindi reveals systematic challenges at the subnarrative level:

- Best performance: **URW: Amplifying war-related fears: Real possibility of nuclear war** (F1 = 1.000)
- **CC: Amplifying Climate Fears: Amplifying existing fears**: F1 = 0.857
- Significant performance degradation on most categories, with many achieving F1 < 0.500

6.4.8 Comparative analysis with baseline architectures

All three actor-critique configurations demonstrate substantial improvements over the transformer baseline, with optimal validation strategy selection depending on specific language and performance requirements:

Hierarchical consistency marginally improved despite performance degradation. While validation strategies generally reduce absolute performance, they do provide slight improvements in hierarchical consistency: no validation shows a 19.4% performance gap between levels (0.6019 vs 0.4077), narrative validation reduces this to 18.6% (0.5919 vs 0.4055), and subnarrative validation achieves 16.8% (0.5758 vs 0.4102). However, this marginal consistency improvement comes at the cost of overall performance reduction, questioning whether hierarchical alignment justifies the performance trade-off.

No-validation configuration achieves optimal overall performance. Contrary to expectations, the simplest configuration without validation delivers the highest average performance across both hierarchical levels (F1 Coarse: 0.6019, F1 Sample: 0.4077). Portuguese exemplifies this pattern most clearly, achieving its peak narrative performance (0.7489) without validation, with both validation approaches causing substantial degradation. This suggests that Gemini 2.5 Flash's base predictions are often more reliable than validation-corrected outputs.

Validation mechanisms introduce systematic over-correction bias. The consistent performance degradation across most language-validation combinations indicates that the critique agents may be systematically over-correcting the base model's predictions. This is particularly evident in Portuguese, where both validation strategies cause significant performance drops (-0.0154 to -0.0585), and in Hindi, where narrative validation severely damages narrative classification (-0.0498) despite the language having the lowest baseline performance and theoretically the most room for improvement.

6.4.9 Key insights and architectural implications

Simplicity outperforms complexity in multilingual hierarchical classification. Our comprehensive ablation study provides strong evidence that the no-validation configuration is optimal for most deployment scenarios. The base Gemini 2.5 Flash model demonstrates superior performance without additional validation layers, achieving the highest average scores across both hierarchical levels while requiring minimal computational overhead. This challenges the assumption that multi-agent validation architectures inherently improve classification accuracy.

Validation mechanisms introduce systematic degradation rather than improvement. The consistent performance reductions across most language-validation combinations reveal that critique agents may be systematically interfering with accurate base predictions rather than correcting errors. The validation process appears to introduce noise and overcorrection bias, particularly problematic for high-performing languages like Portuguese where validation causes substantial performance drops despite strong baseline accuracy.

Language-specific validation effects are largely negative with limited exceptions. Russian represents the only clear case where narrative validation provides consistent benefits (+0.0122 F1 Coarse, +0.0032 F1 Sample), while English shows mixed results with validation type-dependent trade-offs. All other languages experience predominantly negative validation effects, indicating that validation mechanisms may be poorly calibrated for diverse linguistic and cultural contexts in multilingual propaganda detection.

Hierarchical consistency improvements cannot justify performance costs. While validation approaches provide marginal improvements in hierarchical consistency (reducing performance gaps by 0.8-2.6%), these gains come at significant absolute performance costs. The trade-off between hierarchical alignment and overall accuracy strongly favors the simpler no-validation approach, particularly for production systems where classification accuracy is paramount.

Actor-critique architecture effectiveness independent of validation complexity. The substantial improvements over transformer baselines are achieved primarily through the base actor-critique framework rather than validation enhancements. This suggests that the core architectural innovation lies in the prompt engineering and task decomposition rather than multi-agent validation, supporting simpler deployment strategies that maximize both performance and computational efficiency.

The ablation study conclusively demonstrates that validation mechanisms in multilingual hierarchical classification may be counterproductive, with the no-validation actor-critique pipeline providing optimal performance across languages while maintaining architectural simplicity. This finding has significant implications for production deployment strategies, suggesting that computational resources are better allocated to improving base model performance rather than implementing complex validation layers that consistently degrade accuracy.

6.4.10 Cross-architectural comparison: AutoGen vs. Actor-Critique

To understand the relative strengths of our two architectures, we compare the performance of Architecture 1 (AutoGen framework) with Architecture 2 (Actor-Critique pipeline) across available languages and metrics.

Table 6.7: Performance comparison between Architecture 1 (AutoGen) and Architecture 2 (Actor-Critique, no validation) across languages. Architecture 2 shows substantial improvements, particularly in multilingual settings.

Language	Architecture	F1 Coarse/Macro	F1 Sample	Multilingual Approach
EN	AutoGen	0.513	0.406	Native
	Actor-Critique	0.513	0.382	Native
	Δ	**0.000**	**-0.024**	—
PT	AutoGen	0.285	0.173	Translate-to-English
	Actor-Critique	0.749	0.474	Direct multilingual
	Δ	**+0.464**	**+0.301**	—
RU	AutoGen	0.247	0.137	Translate-to-English
	Actor-Critique	0.696	0.475	Direct multilingual
	Δ	**+0.449**	**+0.338**	—

Native English performance parity with different strengths. Both architectures achieve identical narrative-level performance on English (F1 Coarse/Macro: 0.513), demonstrating that the core classification capability is comparable. The Actor-Critique pipeline shows a slight decrease in subnarrative performance (-0.024 F1 Sample), suggesting that the single-agent approach may have minor advantages for fine-grained English classification, possibly due to reduced complexity in the reasoning process.

Dramatic multilingual superiority of Actor-Critique architecture. The Actor-Critique pipeline demonstrates substantial improvements in multilingual settings: Portuguese shows gains of +0.464 F1 Coarse and +0.301 F1 Sample, while Russian achieves +0.449 F1 Coarse and +0.338 F1 Sample improvements. These massive performance gains (163-246% improvements) highlight the critical advantage of direct multilingual processing over the translate-to-English approach used by AutoGen.

Hierarchical consistency advantages in multilingual contexts. While AutoGen shows severe hierarchical degradation in multilingual settings (Portuguese: 0.285→0.173, -39% drop; Russian: 0.247→0.137, -45% drop), the Actor-Critique pipeline maintains more consistent hierarchical performance across languages. The multilingual hierarchical gap averages 36.8% for Actor-Critique compared to AutoGen's 42% average drop, suggesting better error propagation management.

Architectural design implications for multilingual deployment. The comparison reveals that Architecture 2's success stems from two key design decisions: (1) direct multilingual processing eliminates translation-induced information loss, and (2) the unified agent approach reduces complexity-related error accumulation in cross-lingual settings. This suggests that for multilingual propaganda detection, architectural simplicity combined with native language processing significantly outweighs the theoretical benefits of specialized agent decomposition.

6.4.11 Evaluation limitations and dataset size considerations

Limited development set size constrains architectural evaluation robustness. The comparative analysis presented above should be interpreted with caution due to the relatively small size of the multilingual development set. With an average of 32-41 documents per language (BG: 35, EN: 41, HI: 35, PT: 35, RU: 32), the evaluation dataset provides limited statistical power for drawing definitive conclusions about architectural superiority across diverse linguistic and cultural contexts.

Statistical significance challenges for cross-architectural comparison. The substantial performance differences observed between architectures (e.g., +0.464 F1 improvements for Portuguese) may be influenced by the small sample sizes and specific document characteristics within each language subset. While the magnitude of improvements suggests genuine architectural advantages, larger-scale evaluation would be necessary to establish statistical significance and generalizability across broader document populations and propaganda narrative distributions.

Document-level variance impact on performance estimates. The limited document count per language (30-40 documents) means that individual high-performing or challenging documents can disproportionately influence overall performance metrics. This is particularly relevant for rare narrative categories where only a few positive examples exist across the entire development set, making architectural comparison sensitive to document selection effects rather than fundamental design differences.

6.4.12 SemEval-2025 Task 10 competition performance

To contextualize our architectural evaluation within the broader research community, we present our team's (INSALyon2) official ranking in the SemEval-2025 Task 10 competition for Subtask 2 (narrative classification) across all evaluated languages.

Table 6.8: INSALyon2 official rankings in SemEval-2025 Task 10 Subtask 2 across all languages. Rankings reflect performance on the official test set using our Actor-Critique architecture (no validation configuration).

Language	Rank	F1 Macro Coarse	F1 Samples
English	2	0.547	0.433
Portuguese	2	0.679	0.433
Hindi	3	0.515	0.435
Russian	4	0.589	0.410
Bulgarian	5	0.590	0.381

Strong competitive performance validates architectural design. Our Actor-Critique architecture achieved consistent top-5 rankings across all languages, with particularly strong second-place positions in English and Portuguese, validating the effectiveness of our no-validation configuration in multilingual contexts.

Performance consistency confirms development findings. The competition results align well with our development set analysis, particularly the strong Portuguese performance (0.679 test vs 0.749 dev) and the notable Hindi improvement from development challenges to competitive third place, demonstrating good generalization properties of the direct multilingual processing approach.

Comparative analysis with top-performing approaches. While our approach achieved competitive second-place performance in English, the first-place GateNLP team [SSB25a] reported superior results using their hierarchical three-step prompting methodology. Despite implementing their described architectural framework based on their published approach, we were unable to replicate their reported performance levels, suggesting that implementation details, hyperparameter configurations, or specific prompt engineering techniques may play critical roles in achieving optimal results with hierarchical prompting strategies.

7 Conclusion

This paper charted a deliberate course from traditional supervised methods to the frontier of zero-shot agentic reasoning for hierarchical multi-label classification. Our journey began with a state-of-the-art mDeBERTa baseline, which, despite architectural enhancements, confirmed the inherent limitations of fine-tuning on data-scarce, long-tailed tasks. We then demonstrated the viability of zero-shot classification through two distinct LLM-based architectures: a decomposed, multi-agent framework and a holistic, evidence-grounded pipeline.

7.1 Principal Findings and Implications

From this comparative analysis, we draw three principal conclusions that carry significant implications for the field.

1. **The necessity of zero-shot reasoning.** For nuanced, low-resource HMLC tasks like narrative detection, the paradigm of zero-shot reasoning is not merely an alternative but a necessity. LLMs' ability to classify from semantic definitions, rather than statistical patterns in sparse data, offers a decisive advantage that supervised methods struggle to overcome.

2. **The imperative of native-language processing.** For multilingual applications, direct, native-language processing is non-negotiable. Our experiments unequivocally show that the common "translate-then-classify" shortcut introduces unacceptable information loss, destroying the cultural and linguistic nuances critical for accurate analysis.

3. **The value of traceable simplicity.** In the design of agentic systems, architectural simplicity and inherent traceability can be more effective than complex, multi-step validation loops. The superior performance of our single-pass, evidence-grounded Actor over the full Actor-Critique pipeline serves as a crucial finding: forcing a model to

ground its claims in evidence provides more value than adding a complex, and potentially counter-productive, validation layer.

These conclusions directly inform the critical challenges that lie ahead, which will shape the next phase of research in narrative detection and HMLC.

7.2 Key Open Challenges

- **Interpretability and faithfulness**
  - Models must provide human-understandable explanations for predictions and demonstrate structural faithfulness (e.g., obey the true-path rule). Our evidence-grounded approach is a step in this direction, but more formal guarantees are needed.
  - Beyond loss-based penalties, we need architectures and verification methods that certify hierarchical consistency.
- **Dynamic and evolving hierarchies**
  - Real-world taxonomies change: new labels appear, labels split/merge, and relations shift.
  - Research directions: continual learning for label updates, modular label adapters, and lightweight schema evolution strategies that avoid full retraining.
- **Robust cross-lingual and cross-cultural transfer**
  - As our results show, transfer across typologically distant languages and cultural framings remains unreliable.
  - Promising approaches: culture-aware adapters, contrastive cross-cultural fine-tuning, and evaluation protocols that measure cultural equivalence rather than literal translation.
- **Modeling richer narrative structure**
  - Narratives include causality, temporality, and actor-event relations that are not captured by simple taxonomies.

  - Move toward graph-structured prediction, joint event-role-narrative models, and hybrid symbolic-neural representations.

- **Governance of LLM augmentation**
  - LLM-generated training data is a powerful but noisy resource: it may amplify biases or introduce stylistic artifacts.
  - Needed tools: sampling and diversity controls, human-in-the-loop filters, bias audits, and provenance tracking for synthetic examples.

7.3 Concluding Remarks

The shift from fine-tuning to zero-shot reasoning is more than a change in technique; it is a fundamental re-evaluation of how we approach complex classification. Our findings illustrate that future progress in HMLC will depend less on architectural tweaks to supervised models and more on developing robust, traceable, and culturally-aware reasoning systems. Addressing the open challenges of interpretability, dynamic hierarchies, and the responsible governance of synthetic data is therefore not an incremental step, but the central task for building narrative analysis tools that are both powerful and trustworthy in their real-world application. Progress will require integrated work across evaluation, modelling, data governance, and tooling—a multidisciplinary agenda that is both technically challenging and societally important.

Bibliography

[CSB25] Jan Cegin, Jakub Simko, and Peter Brusilovsky. “LLMs vs Established Text Augmentation Techniques for Classification: When do the Benefits Outweight the Costs?” In: *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Ed. by Luis Chiruzzo, Alan Ritter, and Lu Wang. Albuquerque, New Mexico: Association for Computational Linguistics, Apr. 2025, pp. 10476–10496. ISBN: 979-8-89176-189-6. DOI: 10.18653/v1/2025.naacl-long.526. URL: https://aclanthology.org/2025.naacl-long.526/ (cit. on pp. 10, 13).

[CMM11] E. A. Cherman, M. C. Monard, and J. Metz. “Multi-label Problem Transformation Methods: a Case Study”. In: *CLEI Electronic Journal* 14.1 (Apr. 2011). DOI: 10.19153/cleiej.14.1.4. URL: https://doi.org/10.19153/cleiej.14.1.4 (cit. on p. 5).

[Dev+19] Jacob Devlin et al. “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”. en. In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. Minneapolis, Minnesota: Association for Computational Linguistics, 2019, pp. 4171–4186. DOI: 10.18653/v1/N19-1423. URL: http://aclweb.org/anthology/N19-1423 (cit. on p. 2).

[EN25] Mohamed - Nour Eljadiri and Diana Nurbakova. “Team INSALyon2 at SemEval-2025 Task 10: A Zero-shot Agentic Approach to Text Classification”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. Ed. by Sara Rosenthal et al. Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 965–980. ISBN: 979-8-89176-273-2. URL: https://aclanthology.org/2025.semeval-1.129/ (cit. on pp. 10, 29).

[Fan+25] Enfa Fane et al. *Fane at SemEval-2025 Task 10: Zero-Shot Entity Framing with Large Language Models*. arXiv preprint arXiv:2504.20469. 2025. URL: https://arxiv.org/abs/2504.20469 (cit. on pp. 15–17).

[F25] M. Fenu and ... "DEMON at SemEval-2025 Task 10: Fine-tuning LLaMA-3 for Multilingual Entity Framing". In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. 2025. URL: https://aclanthology.org/2025.semeval-1.192/ (cit. on pp. 15, 16).

[GZ24] Anna V. Glazkova and Olga V. Zakharova. *Evaluating LLM Prompts for Data Augmentation in Multi-label Classification of Ecological Texts*. 2024. arXiv: 2411.14896 [cs.CL]. URL: https://arxiv.org/abs/2411.14896 (cit. on p. 10).

[Gon+20] Jibing Gong et al. "Hierarchical Graph Transformer-Based Deep Learning Model for Large-Scale Multi-Label Text Classification". In: *IEEE Access* 8 (2020), pp. 30885–30896. DOI: 10.1109/ACCESS.2020.2973121 (cit. on p. 2).

[Goo24] Google Research. *Introducing Gemini*. https://blog.google/technology/ai/introducing-gemini/. Google Research blog post; accessed 2025-09-24. 2024 (cit. on p. 53).

[He+21] Pengcheng He et al. "DeBERTa: Decoding-enhanced BERT with Disentangled Attention". In: *International Conference on Learning Representations*. 2021. URL: https://openreview.net/forum?id=XPZIaotutsD (cit. on pp. 22, 23).

[Hu+25] W. Hu et al. "A Survey of Multi-Label Text Classification Under Few-Shot Scenarios". In: *Applied Sciences* 15 (2025), p. 8872. DOI: 10.3390/app15168872. URL: https://doi.org/10.3390/app15168872 (cit. on p. 1).

[Hua+24] S. Huang et al. "Application of Label Correlation in Multi-Label Classification: A Survey". In: *Applied Sciences* 14 (2024), p. 9034. DOI: 10.3390/app14199034. URL: https://doi.org/10.3390/app14199034 (cit. on p. 1).

[Hua+21] Yi Huang et al. "Balancing Methods for Multi-label Text Classification with Long-Tailed Class Distribution". In: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Ed. by Marie-Francine Moens et al. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 8153–8161. DOI: 10.18653/v1/2021.emnlp-main.643. URL: https://aclanthology.org/2021.emnlp-main.643/ (cit. on pp. 11, 12).

[Lan24a] LangChain AI. *LangGraph: Build resilient language agents as graphs*. Accessed: 2025-09-19. 2024. URL: https://github.com/langchain-ai/langgraph (cit. on pp. 34, 40).

[Lan24b] LangChain AI. *LangSmith: Developer platform for building reliable LLM applications*. Accessed: 2025-09-19. 2024. URL: https://www.langchain.com/langsmith (cit. on pp. 37, 40).

[Li+23] Dan Li et al. "Enhancing Extreme Multi-Label Text Classification: Addressing Challenges in Model, Data, and Evaluation". In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Ed. by Mingxuan Wang and Imed Zitouni. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 313–321. DOI: 10.18653/v1/2023.emnlp-industry.30. URL: https://aclanthology.org/2023.emnlp-industry.30/ (cit. on pp. 11, 12).

[Li+24] Dongyuan Li et al. *A Survey on Deep Active Learning: Recent Advances and New Frontiers*. 2024. arXiv: 2405.00334v2 [cs.LG]. URL: https://arxiv.org/abs/2405.00334v2 (cit. on p. 12).

[Li+21] Irene Li et al. "Heterogeneous Graph Neural Networks for Multi-label Text Classification". In: *Lecture Notes in Computer Science*. arXiv:2103.14620. 2021. DOI: 10.48550/ARXIV.2103.14620. URL: https://arxiv.org/abs/2103.14620 (cit. on p. 8).

[Liu+23] Rundong Liu et al. *Recent Advances in Hierarchical Multi-label Text Classification: A Survey*. 2023. arXiv: 2307.16265 [cs.CL]. URL: https://arxiv.org/abs/2307.16265 (cit. on pp. 1, 2).

[Liu+24] Zequan Liu et al. "ALoRA: Allocating Low-Rank Adaptation for Fine-tuning Large Language Models". In: *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Long Papers*. Mexico City, Mexico, June 2024, pp. 622–641. DOI: 10.18653/v1/2024.naacl-long.35. URL: https://aclanthology.org/2024.naacl-long.35 (cit. on p. 12).

[Luq+19] Amalia Luque et al. "The impact of class imbalance in classification performance metrics based on the binary confusion matrix". In: *Pattern Recognit.* 91 (2019), pp. 216–231. DOI: 10.1016/J.PATCOG.2019.02.023 (cit. on p. 11).

[Nak+25] Preslav Nakov et al. “BERTastic at SemEval-2025 Task 10: Fine-tuned Transformers for Multilingual Narrative Analysis”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. 2025. URL: https://aclanthology.org/2025.semeval-1.177/ (cit. on p. 16).

[Pen+21] Hao Peng et al. “Hierarchical Taxonomy-Aware and Attentional Graph Capsule RCNNs for Large-Scale Multi-Label Text Classification”. In: *IEEE Transactions on Knowledge and Data Engineering* 33.6 (2021), pp. 2505–2519. DOI: 10.1109/TKDE.2019.2956257 (cit. on p. 2).

[PM25] Jakub Piskorski, Tarek Mahmoud, and ... “SemEval-2025 Task 10: Multilingual Characterization and Extraction of Narratives from Online News”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. ACL Anthology. 2025. URL: https://aclanthology.org/2025.semeval-1.331/ (cit. on pp. 15–18).

[Pis+25] Jakub Piskorski et al. “SemEval-2025 Task 10: Multilingual Characterization and Extraction of Narratives from Online News”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval 2025)*. Vienna, Austria, 2025 (cit. on p. 1).

[P25] P. Premnath and ... “TECHSSN at SemEval-2025 Task 10: A Comparative Study for Narrative Extraction”. In: *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. 2025. URL: https://aclanthology.org/2025.semeval-1.286/ (cit. on p. 16).

[Raz+24] Olesya Razuvayevskaya et al. “Comparison between parameter-efficient techniques and full fine-tuning: A case study on multilingual news article classification”. In: *PLOS ONE* 19.5 (2024), e0301738. DOI: 10.1371/journal.pone.0301738. URL: https://doi.org/10.1371/journal.pone.0301738 (cit. on p. 12).

[Rea+11] J. Read et al. “Classifier Chains for Multi-label Classification”. In: *Machine Learning* 85.3 (2011), pp. 333–359. DOI: 10.7007/s10994-011-5256-5. URL: https://doi.org/10.1007/s10994-011-5256-5 (cit. on p. 5).

[Rea+21] J. Read et al. “Classifier Chains: A Review and Perspectives”. In: *Journal of Artificial Intelligence Research* 70 (2021), pp. 683–718. DOI: 10.1613/jair.1.12376. URL: https://doi.org/10.1613/jair.1.12376 (cit. on p. 5).

[RFR22] Miguel Romero, Jorge Finke, and Camilo Rocha. "A top-down supervised learning approach to hierarchical multi-label classification in networks". In: *Applied Network Science* 7.1 (2022), p. 8. DOI: 10.1007/s41109-022-00445-3. URL: https://doi.org/10.1007/s41109-022-00445-3 (cit. on p. 7).

[RN25] Egil Rønningstad and Gaurav Negi. *LTG at SemEval-2025 Task 10: Optimizing Context for Classification of Narrative Roles*. arXiv preprint arXiv:2506.05976. 2025. URL: https://arxiv.org/abs/2506.05976 (cit. on pp. 16, 17).

[SC22] Mobashir Sadat and Cornelia Caragea. "Hierarchical Multi-Label Classification of Scientific Documents". en. In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, 2022, pp. 8923–8937. DOI: 10.18653/v1/2022.emnlp-main.610. URL: https://aclanthology.org/2022.emnlp-main.610 (cit. on p. 2).

[Sem25a] SemEval-2025 Task 10 Organizers. *SemEval 2025 Task 10 Leaderboard (official results)*. https://propaganda.math.unipd.it/semeval2025task10/leaderboard.php. Accessed 2025-09-16. 2025 (cit. on p. 17).

[Sem25b] SemEval-2025 Task 10 Organizers. *SemEval 2025 Task 10: Multilingual Characterization and Extraction of Narratives from Online News (task webpage)*. https://propaganda.math.unipd.it/semeval2025task10/. Accessed 2025-09-16. 2025 (cit. on pp. 16, 17).

[Sha+18] Jincheng Shan et al. "Co-learning Binary Classifiers for LP-Based Multi-label Classification". en. In: *Intelligence Science and Big Data Engineering*. Ed. by Yuxin Peng et al. Vol. 11266. Cham: Springer International Publishing, 2018, pp. 443–453. ISBN: 9783030026974 9783030026981. DOI: 10.1007/978-3-030-02698-1_39. URL: https://link.springer.com/10.1007/978-3-030-02698-1_39 (cit. on p. 2).

[SSB25a] Iknoor Singh, Carolina Scarton, and Kalina Bontcheva. *GateNLP at SemEval-2025 Task 10: Hierarchical Three-Step Prompting for Multilingual Narrative Classification*. arXiv preprint arXiv:2505.22867. 2025. URL: https://arxiv.org/abs/2505.22867 (cit. on pp. 2, 15–17, 65).

[SSB25b] Iknoor Singh, Carolina Scarton, and Kalina Bontcheva. *GateNLP H3Prompt System Code Repository*. https://github.com/GateNLP/semeval2025-task10. GitHub repository for hierarchical three-step prompting system. 2025 (cit. on p. 2).

[Su+24] Tong Su et al. "Unlocking Parameter-Efficient Fine-Tuning for Low-Resource Language Translation". In: *Findings of the Association for Computational Linguistics: NAACL 2024*. Ed. by Kevin Duh, Helena Gomez, and Steven Bethard. Mexico City, Mexico: Association for Computational Linguistics, June 2024, pp. 4217–4225. DOI: 10.18653/v1/2024.findings-naacl.263. URL: https://aclanthology.org/2024.findings-naacl.263/ (cit. on p. 12).

[Suc+14] L. Sucar et al. "Multi-label classification with Bayesian network-based chain classifiers". In: *Pattern Recognit. Lett.* 41 (2014), pp. 14–22. DOI: 10.1016/J.PATREC.2013.11.007 (cit. on pp. 4, 5).

[Tha+20] F. Thabtah et al. "Data imbalance in classification: Experimental evaluation". In: *Inf. Sci.* 513 (2020), pp. 429–441. DOI: 10.1016/j.ins.2019.11.004 (cit. on p. 11).

[TS18] Vaishali S. Tidake and Shirish S. Sane. "Multi-label Classification: a survey". In: *International Journal of Engineering & Technology* 7.4.19 (2018), p. 1045. DOI: 10.14419/ijet.v7i4.19.28284. URL: https://doi.org/10.14419/ijet.v7i4.19.28284 (cit. on pp. 1, 2).

[Vas+17] Ashish Vaswani et al. "Attention is all you need". In: *Advances in Neural Information Processing Systems*. 2017, pp. 5998–6008 (cit. on p. 22).

[Wan+23] Fengjun Wang et al. "Text2Topic: Multi-Label Text Classification System for Efficient Topic Detection in User Generated Content with Zero-Shot Capabilities". In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Ed. by Mingxuan Wang and Imed Zitouni. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 93–103. DOI: 10.18653/v1/2023.emnlp-industry.10. URL: https://aclanthology.org/2023.emnlp-industry.10/ (cit. on p. 9).

[WDH24] Kunze Wang, Yihao Ding, and Soyeon Caren Han. *Graph Neural Networks for Text Classification: A Survey*. 2024. DOI: https://doi.org/10.1007/s10462-024-10808-0. arXiv: 2304.11534 [cs.CL]. URL: https://arxiv.org/abs/2304.11534 (cit. on p. 8).

[WCB18] Jonatas Wehrmann, Ricardo Cerri, and Rodrigo Barros. "Hierarchical Multi-Label Classification Networks". In: *Proceedings of the 35th International Conference on Machine Learning*. Ed. by Jennifer Dy and Andreas Krause. Vol. 80. Proceedings of Machine Learning Research. PMLR, July 2018, pp. 5075–5084. URL: http://proceedings.mlr.press/v80/wehrmann18a.html (cit. on p. 7).

[Wen+20] Wei Weng et al. “Label Specific Features-Based Classifier Chains for Multi-Label Classification”. In: *IEEE Access* 8 (2020), pp. 51265–51275. ISSN: 2169-3536. DOI: 10.1109/ACCESS.2020.2980551. URL: https://ieeexplore.ieee.org/document/9035463/ (cit. on p. 2).

[WCA23] Chenxi Whitehouse, Monojit Choudhury, and Alham Fikri Aji. *LLM-powered Data Augmentation for Enhanced Cross-lingual Performance*. 2023. arXiv: 2305.14288 [cs.CL]. URL: https://arxiv.org/abs/2305.14288 (cit. on p. 13).

[Wol+20] Thomas Wolf et al. “Transformers: State-of-the-Art Natural Language Processing”. In: *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*. Online: Association for Computational Linguistics, Oct. 2020, pp. 38–45. DOI: 10.18653/v1/2020.emnlp-demos.6. URL: https://www.aclweb.org/anthology/2020.emnlp-demos.6 (cit. on p. 39).

[Wu+24] Qingyun Wu et al. “AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation”. In: *LLM Agents Workshop, ICLR 2024*. Best Paper award. 2024. eprint: arXiv:2308.08155. URL: https://arxiv.org/abs/2308.08155 (cit. on p. 29).

[Xu+21] Linli Xu et al. “Hierarchical Multi-label Text Classification with Horizontal and Vertical Category Correlations”. In: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Ed. by Marie-Francine Moens et al. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 2459–2468. DOI: 10.18653/v1/2021.emnlp-main.190. URL: https://aclanthology.org/2021.emnlp-main.190/ (cit. on p. 6).

[Yan+24] Ruichao Yang et al. “Reinforcement Tuning for Detecting Stances and Debunking Rumors Jointly with Large Language Models”. In: *Findings of the Association for Computational Linguistics: ACL 2024*. Ed. by Lun-Wei Ku, Andre Martins, and Vivek Srikumar. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 13423–13439. DOI: 10.18653/v1/2024.findings-acl.796. URL: https://aclanthology.org/2024.findings-acl.796/ (cit. on p. 10).

[Yua+24] Ling Yuan et al. “Research of Multi-Label Text Classification based on Label Attention and Correlation Networks”. In: *PLOS ONE* 19.9 (2024), e0311305. DOI: 10.1371/journal.pone.0311305. URL: https://doi.org/10.1371/journal.pone.0311305 (cit. on p. 11).

[Zan+24] A. Zangari et al. "Hierarchical Text Classification and Its Foundations: A Review of Current Research". In: *Electronics* 13.7 (2024), p. 1199. DOI: 10.3390/electronics13071199. URL: https://doi.org/10.3390/electronics13071199 (cit. on pp. 1, 6).

[Zha+18a] M.-L. Zhang et al. "Binary relevance for multi-label learning: an overview". In: *Frontiers of Computer Science* 12.2 (2018), pp. 191–202. DOI: 10.1007/s11704-017-7031-7. URL: https://doi.org/10.1007/s11704-017-7031-7 (cit. on p. 5).

[Zha+18b] Min-Ling Zhang et al. "Binary relevance for multi-label learning: an overview". en. In: *Frontiers of Computer Science* 12.2 (Apr. 2018), pp. 191–202. ISSN: 2095-2228, 2095-2236. DOI: 10.1007/s11704-017-7031-7. URL: http://link.springer.com/10.1007/s11704-017-7031-7 (cit. on pp. 2, 4, 29).

[Zho+20] Jie Zhou et al. "Hierarchy-Aware Global Model for Hierarchical Text Classification". In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Ed. by Dan Jurafsky et al. Online: Association for Computational Linguistics, July 2020, pp. 1106–1117. DOI: 10.18653/v1/2020.acl-main.104. URL: https://aclanthology.org/2020.acl-main.104/ (cit. on pp. 7, 8).

Eidesstattliche Erklärung

Hiermit versichere ich, dass ich diese Masterarbeit selbstständig und ohne Benutzung anderer als der angegebenen Quellen und Hilfsmittel angefertigt habe und alle Ausführungen, die wörtlich oder sinngemäß übernommen wurden, als solche gekennzeichnet sind, sowie, dass ich die Masterarbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegt habe.

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